

ELECTRONIC TELEPHONE EXCHANGES

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(The paper was first received 30th October, 1951, and in revised form 8th January, 1952. Proofs were made available to the public 3rd March, 1952, and the paper was read at an EXTRA MEETING of THE INSTITUTION 13th March, 1952.)

SUMMARY

Attention is being given in Britain and other countries to the possibilities of automatic telephone exchanges constructed with electronic switches, in order to determine whether exchanges cheaper and more reliable than the present types using electro-mechanical switches may thereby be produced. The paper discusses some practical forms which electronic exchanges might take, the possibilities and characteristics of such exchanges and—cursorily—the questions of cost and reliability.

Multiplexed, frequency-division and time-division multiplex types of connector switches for speech and line-signal transmission are described, together with typical subscriber's line equipment. General principles of selection systems, with examples of group and free-circuit marking and of a post-setting-up control circuit, are given. Transmission gain and loss, variation of loss, and distortion are features of electronic connector switches not found in existing types using metal-to-metal contacts, but, on the whole, better speech transmission could be given by electronic systems using the connector switches described in the paper, because of the 4-wire switching which would be inherent in most and desirable in all such systems.

Estimates of cost show favourable signs, but are as yet indeterminate. Total system costs must take into account line plant, the design of which may be profoundly influenced by electronic exchanges. Reliability has yet to be proved, but here the signs are also favourable.

(1) INTRODUCTION

The development of automatic telephone exchanges during some six decades has produced a variety of systems, of which about half a dozen have achieved large-scale use in different parts of the world. These systems, so apparently dissimilar in many ways, are nevertheless all derived from the same set of basic principles. A telephone exchange is primarily a switch, which is itself an aggregate of switches, for the interconnection of transmission circuits entering it. One feature common to all existing exchange systems is that the transmission circuits are terminated on switches including metal contacts which are mechanically closed as required through the agency of a control system. The mechanical movement is produced by electro-magnetic attraction of an iron armature which operates through a link system or motor mechanism to bring the contacts together. The control systems are built up largely of the same kind of switches, although other switching devices such as rectifiers and gas-discharge tubes are being used increasingly.

Switches which rely for their action on mechanical movement of metal contacts are subject to wear of the mechanical parts, wear of the contacts if any appreciable sliding motion of one contact over another takes place, erosion of the contacts due to current passed through them, and particularly by current broken by them, and a degree of uncertainty in the electrical contacting due to surface films and dust collected from the atmosphere. Such exchanges thus need a certain amount of continuous maintenance and are subject to some degree of faulty operation. The art of designing automatic telephone-exchange systems is largely that of minimizing the defects enumerated without increasing the cost of the apparatus, and the achievements in this direction have been remarkable. It has been realized for a

number of years, however, that there are switching devices such as vacuum valves, gas-discharge tubes and rectifiers—collectively referred to as electronic switches—which, having no mechanical movement or contacting surfaces exposed to the atmosphere, are free from difficulties due to these causes. They have their own peculiar difficulties, but engineers in many countries have naturally asked whether automatic exchanges can be constructed economically entirely of such devices and, if so, whether further progress towards increased reliability and reduced capital and maintenance costs of telephone exchanges is more likely to be achieved by such means than by further developments of electro-magnetic switches. Recent work in Britain indicates that fully electronic systems which may possibly compete with electro-magnetic systems in cost and reliability can be produced, but the economics and reliability of such systems have yet to be proved. Publication of other work in this and other countries is as yet limited to a few patents, a selection of which is given in the References. These disclosures do little more than suggest specific solutions to the general problem.

The paper describes the first steps which were taken in a research organization confronted with the problems of determining what practical forms electronic exchanges might take, which of these forms was likely to be the most economical and reliable, and how this form might compete with existing electro-magnetic systems. This approach is thought to be consistent with the present state of the art and preferable to describing any one particular system.

The problem of the form an electronic exchange might take is not adequately solved simply by the substitution of electronic switches for electro-mechanical ones in existing systems. An electromagnetic switch is a mechanism which closes metal contacts having characteristics which, in joining two circuits together, are quite different from those of an electronic device such as a valve or rectifier performing the same function. Moreover, the greater operating speed of electronic switches makes possible switching systems which are impracticable with slower-operating electromagnetic switches. The design of electronic exchanges should therefore be approached by way of basic principles rather than analogy with existing exchanges.

(2) CONNECTOR SWITCHES

(2.1) General Requirements

A telephone exchange is a switch for interconnecting transmission circuits, called lines. In the paper it is postulated that an exchange switch is built up of smaller switches assembled in ranks and connected between the ranks with transmission circuits called trunks. Fig. 1(a) shows schematically an N - M unit switch having N -circuits connected on one side and M circuits on the other; when suitably controlled the switch will connect as many as required of the N -circuits each to one of the M -circuits. Connection between an N -circuit and an M -circuit is made by a connector control-unit, CC, in Fig. 1(a). These units may be separate from the N and M circuits as, for example, the cords of a manual switchboard. In automatic systems they are more usually if not invariably provided indi-

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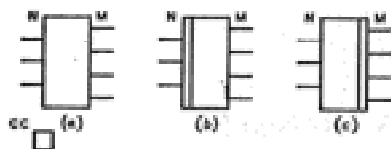


Fig. 1.—Switch symbols.

vidual to either the N or the M circuits, which is represented symbolically by Figs. 1(b) and 1(c) respectively.

Examples of exchanges built up of unit switches in ranks are given in Fig. 2. Fig. 2(a) shows schematically four ranks of

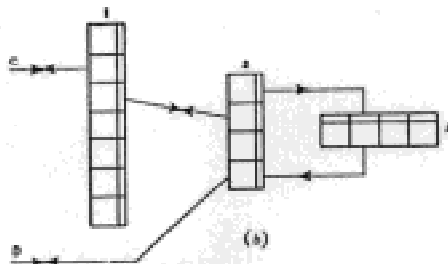
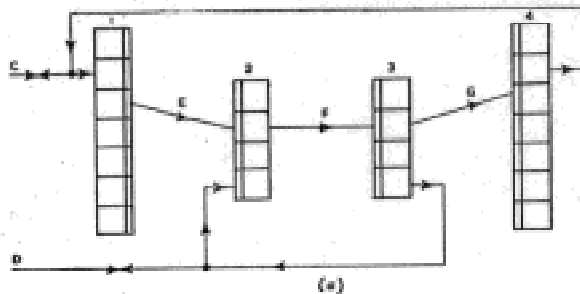


Fig. 2.—Trunking connections.

(a) Two-point connection of lines to exchange.
(b) Single-point connection of lines to exchange.

unit switches, numbered 1 to 4, with two lines C and D connected to the ranks, and trunks E , F and G between the ranks. Many more lines and trunks exist in practice than are actually shown in the diagram. The principles involved in arranging the switches and the connections to them are well known and have been generalized elsewhere.¹⁰ It is sufficient to note here that lines are connected for their calling traffic to the left-hand side of one of the ranks of switches and for their called traffic to the right-hand side of the same or another rank of switches.

The arrangement shown in Fig. 2(a) requires each line to be connected to two points in the exchange. A single-point connection is more convenient and economical in some electronic schemes. Fig. 2(b) shows a trunking arrangement in which the two lines C and D have single-point connections to ranks 1 and 2 respectively for both their outgoing and incoming traffic, the third rank providing a loop connection.

Since the primary function of the switches under consideration is to connect transmission circuits together, they will be called connector switches, and over the connections made by the switches will pass the speech of the users of the system. Bothway speech transmission is, of course, essential. The schematic shows the transmission paths as single lines, but 2-wire, and in some cases 4-wire, transmission circuits are required for speech currents, and still further transmission paths for controlling signals.

The control of the switches and the connections made through them is primarily exercised by line signals, e.g. calling and

answering signals from subscribers' telephone sets. Each line provides transmission paths for both speech and signals, and in a modern system bothway signal transmission is required as well as bothway speech transmission; in fact, more than one signalling channel is often required in one or both directions. The line itself may be a single circuit capable of transmitting all the speech and signals in both directions, or it may be two independent channels, each capable of transmitting the speech and signals in one direction only. A pair of wires, for example, will transmit a.f. currents for speech and d.c., sub-audio and super-audio frequencies for signalling. At some point or points in the system, usually in the exchanges, the speech and signal transmissions have to be segregated into separate paths. The line signals have to be transmitted, in the same way as speech, from one line joined to another through the connector switches in the exchange, and it is natural for the connector switches and trunks to provide connection not only between the speech-transmission paths of lines switched at the exchange but also between the line-signalling transmission paths. It is therefore desirable that the design of connector switches should consider speech and signal transmission together. In Section 2 it will be assumed that, in addition to bothway speech transmission, one signal channel in each direction must be carried through the exchange from line to line connected.

(2.2) Transmission Requirements

The requirements of speech transmission through connector switches are similar to those of transmission lines generally and concern overall loss, attenuation distortion, non-linearity, noise and crosstalk attenuation between circuits passing through the switches. The specification of these quantities in detail is outside the scope of the paper. It is desirable to draw attention to them here, since electronic exchanges will differ from electromagnetic ones in several of these respects. For example, metal-to-metal contacts produce little attenuation, attenuation distortion or non-linearity in transmission. These effects are always present in electronic switches, and their reduction to acceptable values is one of the problems in the design of such switches.

The requirements of signal transmission through connector switches are, in general, similar but less severe than those of speech transmission.

(2.3) Multiplied Element Switches

N - M element switches* can be formed into connector switches of any size, say, xN - yM by multiplying a rectangular array of x by y element switches; the simplest element switches are 1-1 switches such as vacuum valves, rectifiers and gas-discharge tubes.

Figs. 3(a) and 3(b) show respectively vacuum valves and gas-discharge tubes formed into a 3-2 switch. Any n , channel can be connected to any m , channel by a control exerted at the C_{nm} point, the control in Fig. 3(a) being a voltage which causes the grid of the associated valve to pass from the anode current cut-off to the conducting condition, and in Fig. 3(b), to cause the associated gas-discharge tube to conduct. Rectifiers may similarly be used, if a ratio of back/forward a.c. impedance of about 10 000 : 1—demanded by crosstalk attenuation—can be satisfied.

The first point of interest is the numbers of element switches required. The larger the unit switches are made, the greater the numbers of element switches required per switch rank but the fewer the ranks of switches, so that there is an optimum size of unit switch. Very roughly, for 1-1 element switches the optimum size may be taken as 10-10 unit switches, and, averaged

* Element switches are defined in Reference 9 and unit switches in Reference 10.

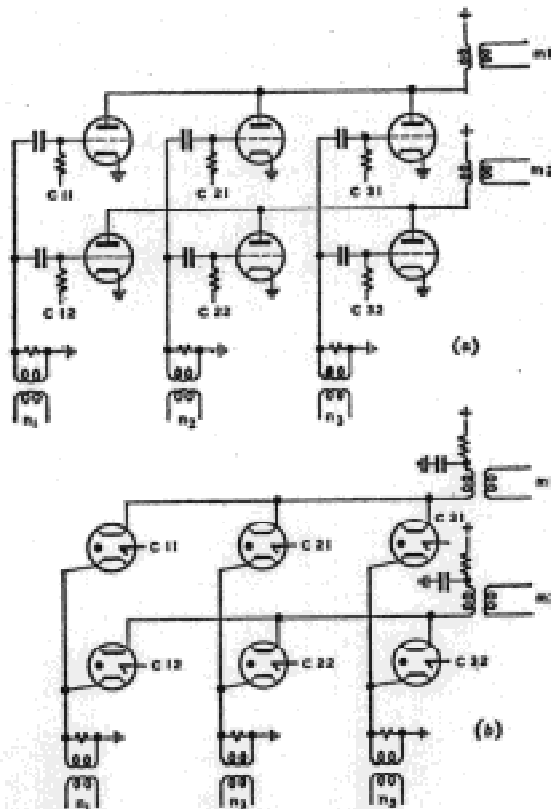


Fig. 3.—Multiplexed element switches.

(a) Vacuum valves.
(b) Gas-discharge tubes.

over the whole exchange, four such switches are needed for each 10 subscribers' lines. Hence 40 element switches per subscriber's line is about the number needed for speech transmission, if the element switches transmit in both directions. Vacuum valves will transmit in one direction only, so that the number stated has to be doubled for bothway transmission and can be ruled out as uneconomic. Gas-discharge tubes provide bothway transmission, although, as mentioned later in this Section, one-way transmission may be essential in some ranks of switches. They consume no power except when conducting; the economics of connector switches constructed with gas-discharge tubes is directly dependent on how cheaply the large numbers required can be made.

If 1- x element switches are used, their numbers relative to 1-1 devices are reduced by a factor x . Beam-switching vacuum valves such as cathode-ray tubes can be made with large values of x , so that they might come within the economic range. Gas-discharge tubes can also be made with the equivalent of 10 or more tubes having a common cathode or anode in a single envelope, which form might be more economic than the equivalent greater number of simpler tubes.

Transmission of speech through vacuum-valve switches using individual valves is satisfactory. The amplification provided by the valves can be reduced by feedback to produce stable zero-loss transmission practically free from distortion. In common with other types of switch, the circuits are electrically unbalanced within the switches; satisfactory crosstalk attenuation is thus dependent on the spacing of conductors and screening.

Speech transmission through gas-discharge-tube or rectifier switches is bound to suffer attenuation which may be assessed as at least 1 db per switching stage. With five to ten switching

stages per exchange, and eight to ten exchanges in the longest calls, the total loss is serious and has to be made up by amplification, either in the exchanges or in the lines between exchanges. The variation in the difference between the loss in the switches and the gain in the amplifiers is liable to be serious. If only 2-wire transmission is provided through the switches, amplification by 2-wire repeaters introduces well-known difficulties. In consequence, 4-wire transmission through at least some of the switches may become necessary, and in these switches the number of unit switches required is doubled. Gas-discharge-tube conduction is liable to introduce noise and attenuation distortion, because a.c. conductance depends on frequency, but tubes in which these effects are negligibly small may be produced.

No simple and inexpensive method of signal transmission through multiplexed element switches of the type under consideration is readily apparent. The change in potential of one of the electrodes of an element switch when it begins to conduct provides a d.c. signal which is not very convenient to use. Alternatively, a.c. transmission demands either separate element switches for signalling or bandwidth outside the speech band over which a.c. signals can be transmitted and separated from the speech by filters.

It was concluded that, although multiplexed element switches, and more particularly those using gas-discharge tubes, may find application in very small exchanges where the features enumerated become less difficult, for larger exchanges the multiplex types of switches described in the following Sections were more attractive.

(2.4) Frequency-Division Multiplex Switches

Carrier-frequency telephony methods may be readily applied to the construction of switches of considerable size. Fig. 4

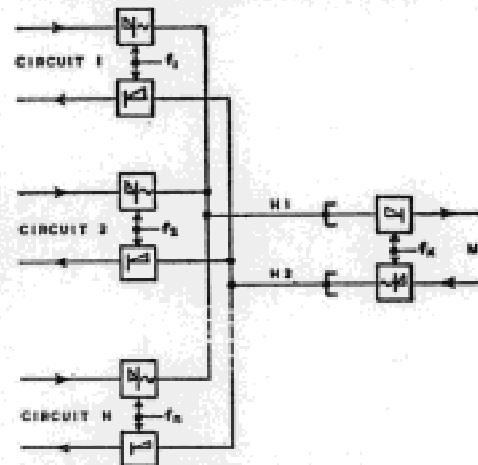


Fig. 4.—Principle of multiplex switch.

shows schematically the principles involved. On the left-hand side of the figure are N a.f. circuits, each consisting of a transmitting and a receiving channel. The transmitting channels each modulate one of N carrier frequencies, the modulated carrier frequencies being applied to a common channel H1. A circuit M has its receiving channel connected to the channel H1 through a demodulator, which is supplied with a demodulating frequency f_c . The transmitting channel of the M -circuit modulates the frequency f_c as a carrier which is applied to a common circuit H2 to which the receiving channels of the N -circuits are connected through demodulators fed with the same frequencies as their respective transmitting modulators. The circuit M is electrically connected to any one of the N -side circuits by making its carrier frequency f_c the same as that of

the carrier frequency of the selected N -side circuit. The arrangement so far described is analogous to a single unselector having a bank of N -fixed contacts and a set of wipers which can be made to contact any of the N -bank contacts. Any number of M -circuits may be commoned to the channels H1 and H2, corresponding to multiplying the banks of unselectors and similar switches. The number of N -circuits may be 100 or more. Switches constructed in this way may be used very much the same as 100-point unselectors in existing systems.

Carrier terminal equipment for transmission over distances is expensive if the best use is made of a line or radio transmission system. For an exchange switch the situation is quite different. The problem is to produce simple and cheap terminal equipment for speech and signal transmission over a negligible length of common circuit. Wide channel spacing, the transmission of carrier, upper and lower sidebands for speech, and the on-off switching of the carrier for signalling are first steps in the solution. Signalling with the carrier wave has some limitations in practice, since speech cannot be transmitted when the carrier is switched off, but is adequate for the basic signals. Other possibilities of economy in design are illustrated by the system⁷ shown schematically in Fig. 5.

carrier frequencies in the range 4 000–6 475 kc/s by 25-kc/s steps. The unwanted modulation products from the outputs of all the second modulators fall outside the range 2 502·5–4 997·5 kc/s, so that all the second modulators can be directly connected to a filter WF3 passing that range of frequencies. The output of the filter is passed via a common amplifier A1 to the receiving channels, which comprise, in series, a modulator FC3 to which one of the carrier frequencies between 4 000 and 6 475 kc/s is applied, a band-pass filter WF4 with its pass-band centred on 1 487·5 kc/s and identical with filter WF2, a detector FC4 consisting of a rectifier, a low-pass filter WF5 for which a resistor shunted by a capacitor suffices, and a channel amplifier A2. The modulator controlled from the signal channel and the detector, low-pass filter and amplifier on the receiving channel [the portion of Fig. 5(a) enclosed by a broken line] is shown in greater detail in Fig. 5(b). The transmitting ring modulator is normally balanced, but is unbalanced by negative current over the signal path. The amplifier valve is normally cut off by bias voltage $-v$ at its grid, and has in its anode circuit a resistor shunted by a capacitor, and the a.f. output transformer. The output from the band-pass filter WF4 is detected by rectifiers MR1 and MR2 and produces a mean positive voltage due to the

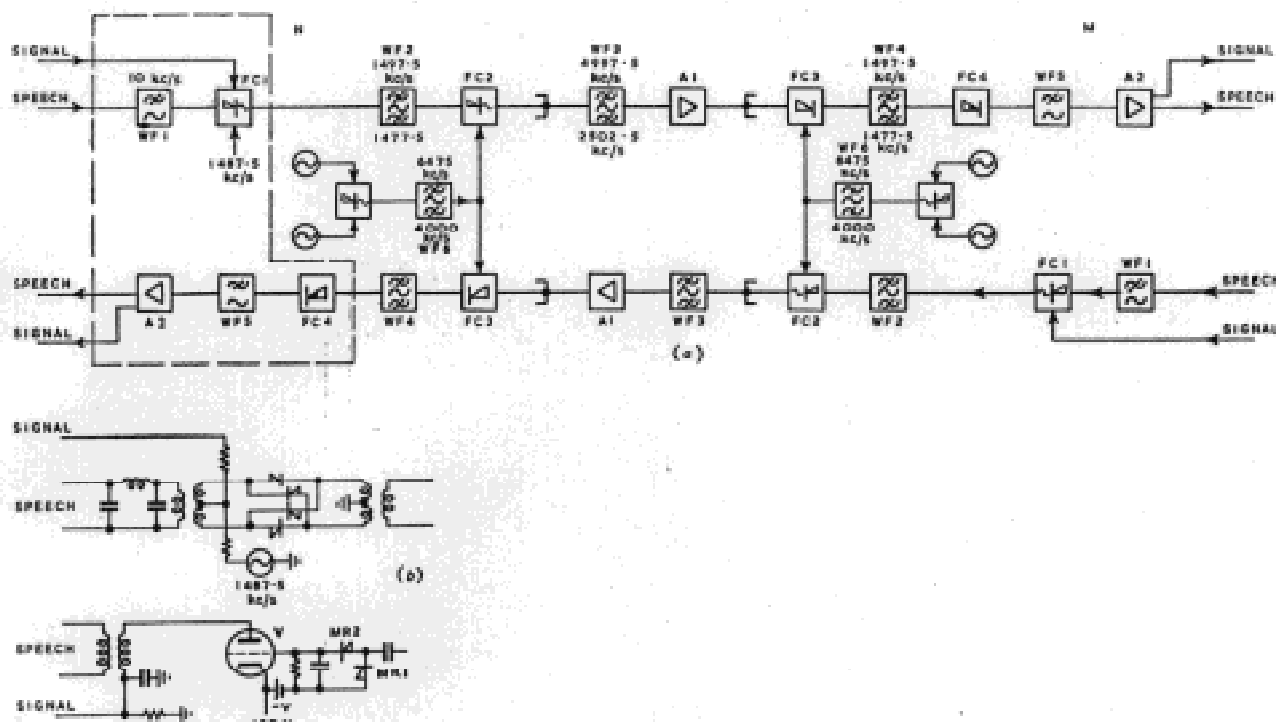


Fig. 5.—Frequency-division multiplex switch.

(a) Symbolic schematic.

(b) Circuit of 4-wire channel ends.

In this system the switching is provided between 4-wire circuits by carrier systems having 100 channels spaced at 25 kc/s between 2 512·5 and 4 987·5 kc/s. The transmitting channel of each circuit has a low-pass filter WF1 cutting off at 10 kc/s, followed by a modulator FC1 supplied with carrier frequency at 1 487·5 kc/s. The modulator is of the ring or other balanced type to which a signal channel is connected and arranged so that the carrier is either suppressed or transmitted according to the signal applied. The modulator is followed by a band-pass filter WF2, which selects the carrier and both adjacent sidebands and is followed by a balanced second modulator FC2 supplied with one of the

transmitted carrier on which the audio frequencies are superimposed. The consequent voltage drop across the anode resistor can be applied directly to the signal path of the next-stage switch.

The method of generating the carrier frequencies required is important in a system of this kind, particularly with regard to the need for selectively switching the carriers for the M -side circuits. In the system shown in Fig. 5, the carrier frequencies from 4 000 to 6 475 kc/s are produced by modulating one frequency out of each of two groups of ten frequencies, the first group together consisting of frequencies between 2 775 and

3 000 kc/s in 25-kc/s steps, and the second of frequencies between 7 000 and 9 250 kc/s in 250-kc/s steps. Each modulator is followed by a band-pass filter WF6 passing 4 000–6 475 kc/s. The N -circuits operate with fixed carrier frequencies. An M -circuit can be connected to any N -circuit by switching the appropriate pair of frequency supplies to the carrier-generating modulator of the M -circuit.

In this system, the filters are simple, and since only three designs account for practically all the filters needed, quantity production is facilitated. Owing to the high frequencies involved, the filter elements are small, and because of the wide frequency spacing, the elements do not need to be very accurate. Each 4-wire terminal requires only one hot-cathode valve [Fig. 5(b)] for both-way speech and one signal path in each direction. Zero-loss transmission of both speech and signals is readily secured and made stable within acceptable limits, and noise, crosstalk and linearity present little difficulty. There is thus good prospect of producing inexpensive equipment which is satisfactory in its transmission properties.

The 4-wire terminals need to be cheap, since every subscriber's line, as well as every junction, terminated on the exchange needs one such equipment; a trunking arrangement as shown in Fig. 2(b) would naturally be chosen. Each trunk within the exchange needs two 4-wire terminals; 100 circuits on the fixed or bank side of the switches ensure economy in the number of trunks needed. Very roughly, the number of trunks will vary from 10 to 100% of the number of lines connected to the exchange, according to the calling rate.

(2.5) Time-Division Multiplex Switches

Fig. 4 may be taken to represent time-division multiplex switches^{2,3,4,5,6} if the modulators and demodulators are replaced by pulse modulators and the carrier frequencies by trains of pulses. The pulse modulators of the N -circuits are then operated by trains of pulses which are fixed in time relative to the multiplex cycle and spaced so that in effect the channels are connected to the common channels H1 and H2 in order. The channels of the M -circuit are operated by a train of pulses which can be selected to coincide in time with the pulse train of any N -circuit. In this way the M -circuit can be connected to any N -circuit, and as many M -circuits as required can be connected to the common channels. Speech and signal transmission can be readily provided through the switch, as can be best explained by describing an example which is illustrated in Fig. 6(a) in some detail. Only one N - and one M -circuit are shown. Taking first the lower half of the diagram and reading from left to right, speech is applied to the input transformer T1, which is terminated in a resistor and a small capacitor C1, and larger capacitor C2. The capacitor C2 is also connected to a signal lead and is normally at earth potential. The capacitor C1 is connected to a modulator including four rectifiers MR1–MR4. The common connection B1 of the four rectifiers has connection through resistors R1 and R2 to positive and negative voltage supplies such that the potential of point B1 is normally positive with respect to earth. Rectifiers MR3 and MR4 are then conducting and rectifier MR1 non-conducting. Rectifier MR2 is connected to a common channel H1 which is also normally at earth potential; this rectifier is also normally non-conducting. Clearly, the speech channel is normally disconnected from the H1 channel, and this is also true of the many similar speech channels in the group commoned to channel H1. To the modulator is applied via a capacitor C3 a train of negative pulses p_x such that each pulse causes the rectifier MR3 to switch to its high-resistance condition. The potential of point B1 then falls until it is clamped via rectifier MR1 to the instantaneous potential of capacitor C1, the channel H1 taking up approxi-

mately the same potential and the rectifier MR4 presenting a high resistance. When the signal lead is at earth potential, the potential of H1 will change very little. If the potential of the signal lead is $-V$ volts, each pulse of the train p_x will cause the potential of the common channel H1 to become $-V$ volts modulated according to the instantaneous voltage across capacitor C1 due to the speech input. Hence, signals are indicated by the mean amplitude of the pulses and speech by the modulation of the mean amplitude. Each channel commoned to H1 has a different pulse-train connected to its capacitor C3, the pulse trains being time spaced, preferably evenly. The circuit to the right-hand side shows one of a second group of channels and includes a modulator with rectifiers MR5 to MR8 corresponding to the rectifiers MR1 to MR4 of the transmitting modulator. It also includes a high-impedance RC combination R5 C5, with a time-constant of about one-half the channel-pulse period, and a valve amplifier. The pulse train p_y is supplied by a channel-pulse generator, CPG, which when suitably controlled provides a pulse train coincident with that of any desired transmitting channel. The valve of the receiving channel receives the modulated channel-pulse which is demodulated back to audio frequency by the low-pass filter WF1 in the anode circuit. The mean amplitude of the pulse is proportional to the component in the anode circuit at the pulse repetition frequency. The first section of the low-pass filter has coupled to it a circuit which rectifies the a.c. component in the anode circuit and applies the d.c. output to a signal lead which can be directly coupled to the input signal lead of the next switching stage. The apparatus in the lower half of Fig. 6(a) provides one-way speech and signal transmission; that in the upper half provides speech and signal transmission in the opposite direction, using the same p_x and p_y supplies. Amplifiers may be necessary in the common channels H1 and H2, but they are not shown in the Figure.

Speech frequencies from 0.3 to over 4 kc/s can be transmitted with a pulse repetition frequency (p.r.f.) of 10 kc/s for the p_x and p_y pulse trains. For 100 channels on the fixed-pulse-time side of the switch, the time spacing between the channels is then 1 microsec, and a pulse width of about 0.3 microsec is desirable. For high crosstalk attenuation¹¹ between the pulses, a band of frequencies from at least 100 c/s to 5 Mc/s has to be transmitted over the H1 channel. The high frequencies involved present some difficulties in the modulators, but germanium-crystal diodes, with their low capacitance and high back resistance are satisfactory in all places except MR5, for which a thermionic diode is necessary.

The pulse supplies for the fixed-time pulses may be generated in various ways, such as frequency division from a master source of pulses, which, for a 100-channel system with a p.r.f. of 10 kc/s, has a pulse period of 1 microsec. The variable-time pulse generators, CPG, may also take various forms, including frequency dividers and timers synchronized from a master source of pulses.

The similarity between frequency-division and time-division switches is readily apparent. The number of 4-wire terminals required is about the same. The somewhat simpler apparatus of the time-division switches suggests that they can be made rather more cheaply than frequency-division switches.

Fig. 6(b) is the schematic equivalent of Fig. 6(a). The fixed-time multiplexes are represented by MX1 and MX2, the arrows showing the directions of transmission. The variable-time multiplexes are shown as modulators RM and TM connected to the common channels, or "highways," H1 and H2. The thickened lines are a.f. transmission circuits, and the signal channels are shown parallel to the speech path they accompany. Pulse channels are shown by broken lines.

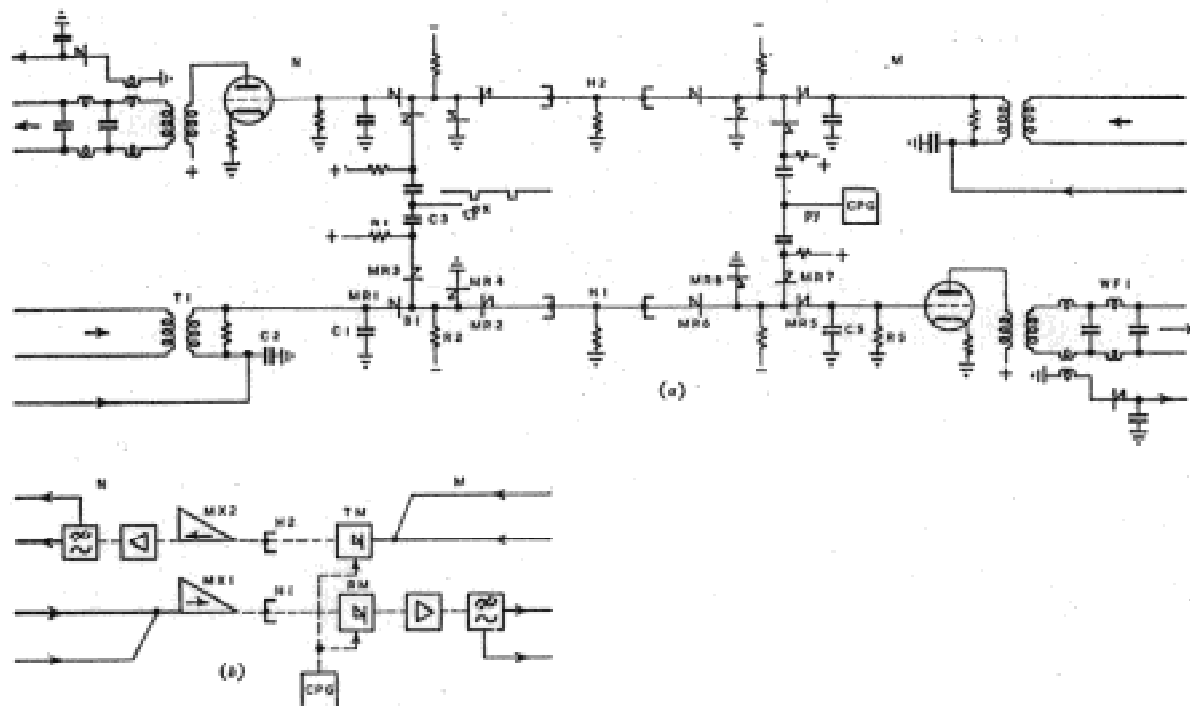


Fig. 6.—Time-division multiplex switch.

(a) Circuit schematic.

(b) Symbolic schematic.

(2.6) Other Forms of Electronic Switch

Electronic exchanges using forms of construction not included in the foregoing Sections have been proposed, notably coaxial-cable networks to which carrier equipment at subscribers' premises is connected.^{1,6} These systems are modifications of the general types included in the paper, and will not be described in detail.

(3) SUBSCRIBER'S LINE EQUIPMENT

From the brief outline of various types of electronic connector switches given in the preceding Sections it is evident that, whatever form of construction is adopted, the switches can be expected to have the capacity to transmit power only of the order of maximum speech power, e.g. one or two milliwatts. This is insufficient for some existing purposes, notably for ringing a conventional type of magneto bell and operating a meter. There are great advantages in retaining the present types of subscribers' station apparatus and exchange meters, and the problem is then presented of providing at the exchange equipment which will successfully link these equipments and the new exchange switching apparatus. Fig. 7 represents an equipment suitable for the frequency-division switch shown in Fig. 5 and the time-division switch shown in Fig. 6.

A conventional telephone is connected by a pair of wires AB to the exchange line-equipment having a hybrid transformer T1, with line balance, for coupling the pair AB to the 4-wire channels of the connector switch. The transmitter feed current is supplied from a central battery and applied to the split 2-wire line winding of transformer T1 through resistors R1 and R2 and the secondary windings of the 17-c/s transformer T2. The subscriber's line signals are derived from the change of potential across resistor R1. In the receive channel are two tuned transformers, T3 and T4, tuned to different frequencies. Transformer T3 is resonant to a signal intended to ring the subscriber's bell and supplied from

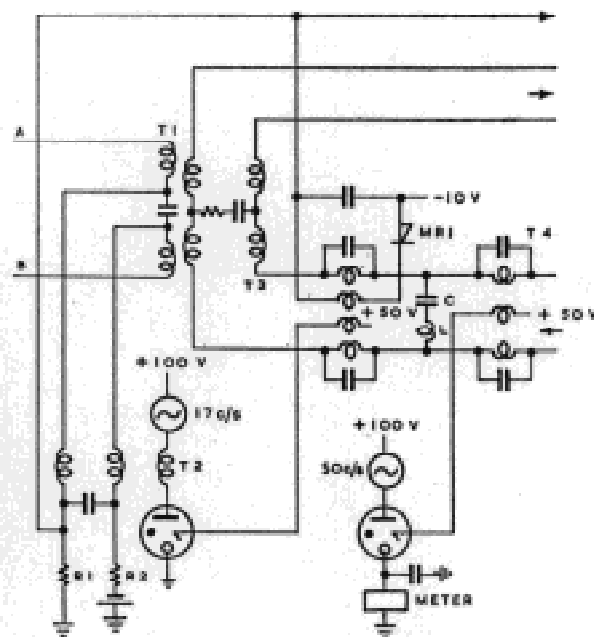


Fig. 7.—Subscriber's line equipment.

exchange equipment connected through the connector switches. The output from the transformer fires a cold-cathode gas-discharge-tube relay in the anode circuit of which is the transformer T2 and a d.c.-biased 17-c/s power supply. When the tube is discharging, sufficient power to ring the bell is supplied to the line. The meter is similarly operated by 50-c/s current through a gas-filled relay operated by the output from transformer T4. Clearly, the gas-discharge tubes must not be

operated by speech currents. The transformer T3 is made ineffective when the line AB is looped by short-circuiting, the change in potential across resistor R1 being used to make rectifier MR1 conduct. The meter circuit is operated by a super-audio-frequency signal which is excluded from the speech-currents by filters in the trunks between switches and locally by a tuned LC circuit.

(4) SELECTING SYSTEMS

The previous Sections have been concerned with the problem of providing connector switches in sufficient size and number, and line equipment such that any line connected to the exchange may be connected through the switches and over the trunks to any other line. The connections made are controlled, directly or indirectly, by the line signals which originate primarily from calling subscriber and junction lines and which are extended through the switches, as has also been described. The line signals involved in the selection process are calling signals and called-number signals. Calling signals are simple, consisting of a single change in the electrical condition of a signal channel. Any selection required as the result of a calling signal is usually called "line finding." The term "selection" is usually applied to selection which results from called-number signals.

Called-number signals may take several forms, such as dial pulses or multi-frequency coded pulses. The actual form is unimportant to the selecting system; essentially, a number of one or more digits is signalled and each digit or combination of digits signifies a group of called circuits, to a free one of which the calling line is required to be connected. The group of called circuits may be a group of trunks, junctions or subscribers' lines, including the case of a one-line subscriber as a group of one circuit. The actual connection made is dependent not only on the calling circuit and the called number, but also on whether there is a free circuit in the group designated by the called-number signals, and the availability of switches and trunks between the calling circuit and the circuits of the called group. Selecting systems in general is too lengthy a subject to be discussed in detail in this paper. Instead, it will be postulated that selection in its simplest form is represented schematically by Fig. 8 and leads to three basic operations, namely group selection, i.e. the selection of a group of called circuits; free-

circuit selection, i.e. the selection of a free circuit in the selected group; and connection, i.e. the connection of the selected called circuit to the calling circuit.

In electro-mechanical systems, both group and free-circuit selection may be accomplished wholly or in part by mechanical motion of a contact relative to circuits arranged in a known order in space. In a wholly electronic system, group selection may be taken to imply the production of an electric signal, called a mark or marking signal, which marks a group of circuits indicated by called-number signals from a calling circuit, and free-circuit selection the similar marking of a free circuit in the selected group.

(4.1) Group Selection

In Fig. 8 an N - M switch has calling circuits on its N side. When an N -circuit signals that it requires a connection, its forward-signal channel is connected to a "marker" apparatus including a group marker, GM. If selector action is required, the called-number signals transmitted from the calling circuit over its forward-signalling path are received into the group marker, which converts them to a mark on a marker lead GM1, GM2 of a group of M circuits designated by the called-number signals. In Fig. 8, the forward-signalling path is connected directly to the group marker. In some systems, as is well known, the forward-signalling path is connected first to a register which receives the called-number signals and retransmits them to the group marker. The number of leads at the output of the group marker varies according to the system in use. It may be as few as ten in a decimal step-by-step selecting system (analogous to the Strowger system), and up to 10 000 or more in a non-decimal system (analogous to the Bell cross-bar system). Electronic group-markers to convert called-number signals of any type to any kind of marking signal on one out of any number up to 10 000 or more marker leads are neither difficult nor expensive to construct. Fig. 9 is an example of apparatus

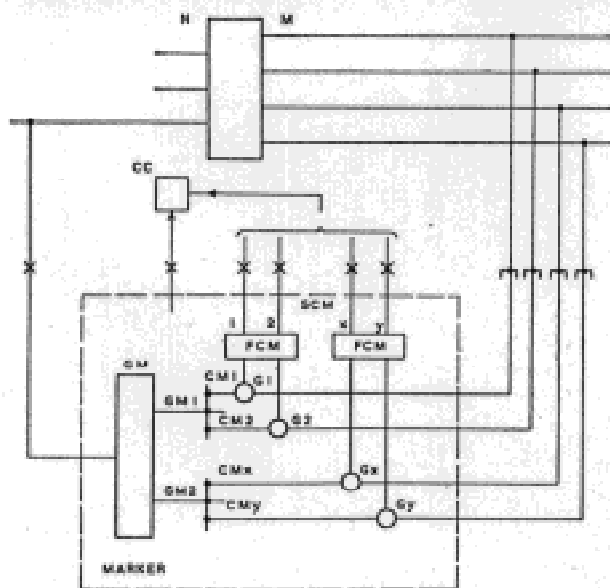


Fig. 8.—Schematic of selection system.

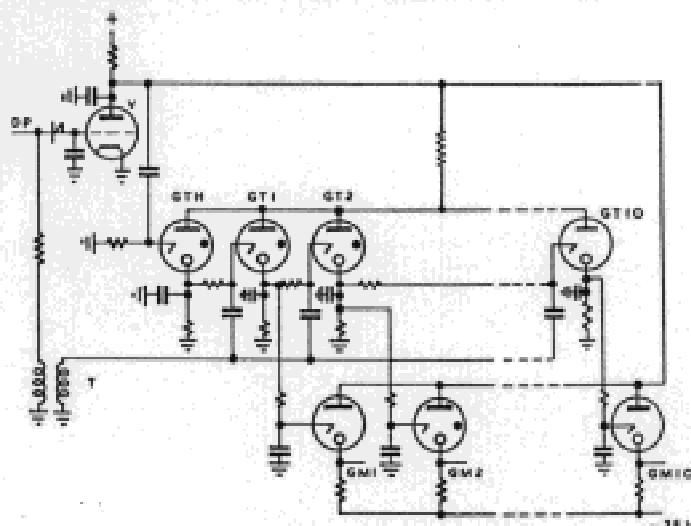


Fig. 9.—Dial-pulse-operated group marker.

designed to convert a train of dial pulses to a marking signal on one of ten marker leads according to the digit dialled. The dial pulses are applied to input lead DP, which is normally at earth potential, but which changes to -10 volts when the marker is seized and returns to earth potential for each break of the dial, finally returning to earth potential to release the marker. These signals may be derived, for example, from the subscriber's line equipment of Fig. 6 directly or via a connector

switch in the usual way. A vacuum triode V has its control grid connected to lead DP through a rectifier and to earth through a capacitor. The capacitor quickly takes up the -10 volts potential of lead DP, but only slowly reaches the potential of that lead when it returns to earth potential. Valve V, which is normally conducting, is therefore cut off quickly by the seizing signal, remains cut off during dialling and re-conducts on release. Its anode circuit controls the anode supply voltage of two chains of cold-cathode gas-discharge tubes. The top row consists of a tube GTH, which strikes when V cuts off, and ten further tubes in a conventional counting chain. Tubes GT1-GT10 strike in that order for each pulse over the lead DP, each pulse being differentiated and voltage amplified by the transformer T. Each tube in the second row is coupled to a tube in the first row through an RC delay circuit of time-constant greater than the period of the dial pulses, so that it is only the final tube discharging in the first row which strikes its tube in the second row. This second-row tube marks its marker lead GM by changing the potential of the lead from -25 to $+25$ volts.

Fig. 10 shows a different kind of marker. Enclosed in the

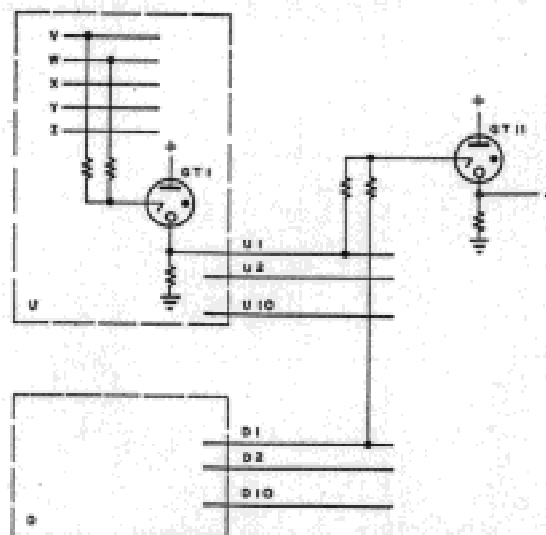


Fig. 10.—Coded-pulse-operated group marker.

rectangle U are five leads lettered V to Z, which are marked by two-out-of-five code from, say, a multi-frequency coded-pulse receiver. Each lead, which is normally at earth potential, is raised when marked to $+80$ volts. Gas-discharge tube GT1 has its primer connected to leads V and W through equal resistors, and fires if both leads are marked. When it fires, it raises its cathode potential, and therefore that of lead U1, to 80 volts above earth. There are nine other tubes connected to leads U2 to U10, and each is operated by one of the combinations of two out of the five leads marked V to Z. The ten leads U1-U10 may be used as group-marker leads; or in combination with ten similar leads, D1-D10, operated by apparatus in rectangle D similar to that in rectangle U, one of 100 tubes such as GT11 may be operated for each combination of one lead marked in each of the groups U1 to U10 and D1 to D10. This arrangement can clearly be extended indefinitely. Where the number of group-marker leads greatly exceeds 10, selection is made through several ranks of switches simultaneously and not through a single *N-M* switch as shown in Fig. 8. Discussion of these more complicated systems is outside the scope of the paper.

The examples given have been chosen to illustrate the kind of operation and apparatus involved. In these examples,

approximately one to two gas-discharge tubes are required per marker lead. There is a more economical arrangement which may be used in appropriate cases, in which for, say, a four-digit numbering scheme, four groups of ten bus-wires are provided, each bus-wire being powered by one valve or gas-discharge tube. A group-marker lead corresponding to a 4-digit number is marked by energizing the appropriate bus-wire in each group; for this purpose the group-marker leads are connected through four resistors to one of the bus-wires in each group and are marked when the bus-wires to which they are connected are energized.

(4.2) Free-Circuit Selection

When the required group of circuits has been selected, the problem is first to determine which circuits are free and then to select one of the free ones. There must therefore be some means of indicating the condition—free or engaged—of each circuit in the group, which for the moment will be assumed to be a signal over a test lead, there being a test lead for each circuit.

Fig. 8 shows the basic conditions as test leads from the *M*-circuits, each test lead signalling the free or engaged condition of its *M*-circuit to a gate circuit G, the gate circuits being joined in groups by circuit-marker leads CM to the group-marker leads. A mark on a group-marker lead passes over all the CM leads joined to it and thence through those gate circuits whose test leads signal the free condition, over free-circuit marking leads to a free-circuit marker FCM, such as that described below and shown in Fig. 11. FCM selects a free

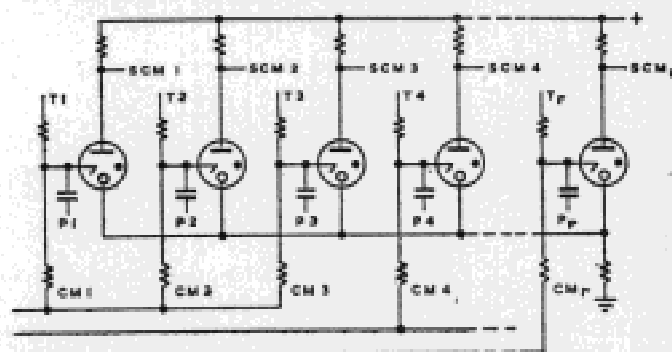


Fig. 11.—Free-circuit selector.

circuit and signals the selected circuit over a selected-circuit marking lead SCM. The reference to leads is symbolic; groups of frequencies or time-spaced trains of pulses can be used for some of the groups of leads.

The marker signals used and the free-circuit-marker apparatus can take many forms. Fig. 11 shows an arrangement using d.c. marker signals and a gas-discharge-tube free-circuit selector. A number of gas-discharge tubes having their cathodes connected to a common resistor can be arranged so that, if one tube fires, the rise in potential of the cathodes will prevent any other tube from firing. This device is used in the free-circuit selector of Fig. 11. A gas-discharge tube is provided for each of the *M*-circuits of Fig. 8, and its primer electrode is controlled from three sources, namely the test lead T for the circuit, a circuit-marker lead CM from the group marker, and, through a capacitor, a pulse source *p*. Positive voltages are used on the test leads to signal the free condition, on the group-marker leads to mark the circuits in groups, and for the pulses. If the cathode of a tube is at earth potential, the sum of the three positive voltages is sufficient to fire the tube. The sum of less than three positive voltages, or the sum of three positive voltages if the cathode potential is raised by any other tube, is insufficient

to fire the tube. With this arrangement, one tube corresponding to a free circuit will fire when a group is marked, the particular one being determined by the order of application of the pulses p_1 to p_r . The selected circuit is marked by an SCM lead by the change of potential of the anode circuit of the conducting tube.

Free-circuit markers of the type shown in Fig. 11 are of general application to any type of connector switch. For example, the d.c. marking on an SCM lead can indicate an element of a multiplied element switch, a frequency in a frequency-division switch or a pulse train in a time-division switch. However, less general types of marker may be more desirable for frequency- and time-division switches, as will become clearer in the next Section.

(4.3) Connection of the Calling Circuit to the Selected Called Circuit

The connection of the calling circuit to the selected called circuit in particular cases is dependent on the type of switch used. The switch with vacuum valves shown in Fig. 3(a) needs a d.c. signal to a control point C_{22} ; for the gas-discharge tubes of Fig. 3(b), a pulse applied to a control point C_{25} suffices; for frequency-division switches like those shown in Fig. 5, steady carrier frequencies have to be switched, and for time-division switches as shown in Fig. 6 a channel-pulse generator has to be set to generate a train of channel pulses at the requisite times in the multiplex cycle. More generally, as shown symbolically in Fig. 8, the marker has to indicate to a connector-control element CC the calling and the selected called circuit to be connected together. The process is relatively simple if each of the N -circuits of Fig. 8 is provided with an individual marker and connector-control element. This is well known in step-by-step systems of selection. It endows exchanges with well-known characteristics of trunking, and other features, the most important of which is that any number of calls can be in process of being set up simultaneously without mutual interference. Electronic exchanges with connector switches of any type can be constructed according to this principle. It is extremely doubtful, however, that any such design would compete in cost and size with existing types of exchange. Economy usually results by making one or more markers common to a number of circuits—the so-called common-control system. This reduces the number of markers needed at the expense of extra switches which must be provided to associate a marker with an N -circuit and connector-control element as required. The switches are interposed in the appropriate leads as indicated by crosses in Fig. 8; the test leads can be commoned as shown. The extra switching is a straightforward problem, although the number of leads involved may make its solution an expensive one. Of greater practical difficulty is the fact that calls which are set up through the exchange are no longer mutually independent. This leads in existing systems to complications which are well known but too involved for discussion in the paper. The difficulty can be avoided, if the switching speed is high enough, by receiving all originating calls into register-senders which store the called-number signals until sufficient have been received completely to determine the whole connection in the exchange, and then controlling the registers so that only one can be sending into the exchange switches at a time. There is then no question of interference between calls being set up simultaneously. Furthermore, only one marker per NM switch is necessary. With electronic switching the switching speed can be high enough. The problem is then reduced to that of connection of the registers and connector switches to the marker. With one marker per NM switch, the connection between the marker and the switch is greatly simplified and largely reduced to commoning. It thus

becomes possible to produce very economical selecting systems for all types of electronic exchange. A selecting system using time-division multiplex pulse-markers for time-division multiplex connector switches is probably the simplest and most economical for a variety of reasons, including that the connection between the marker and the connector control elements needs only two leads. Selectors and line-finders of an exchange system of this type are shown schematically in Figs. 12 and 13.

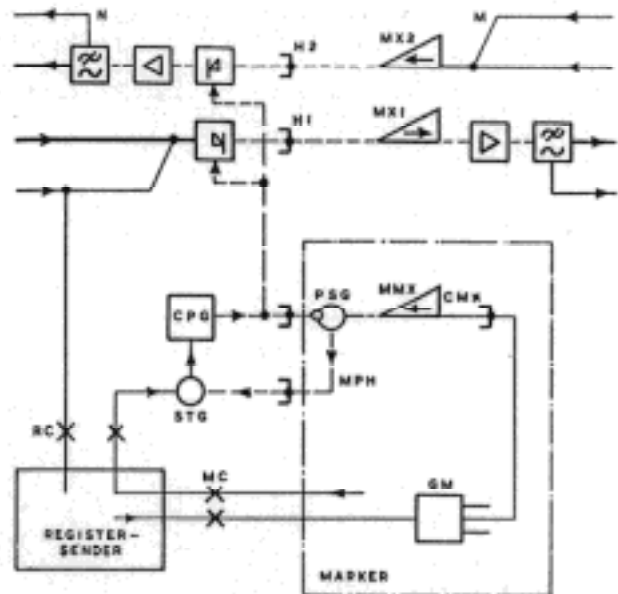


Fig. 12.—Time-division multiplex selector.

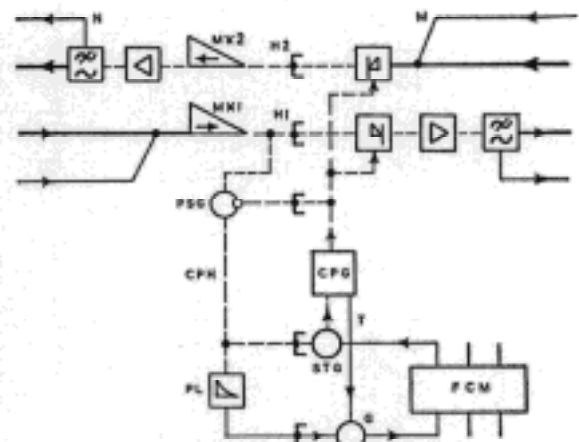


Fig. 13.—Time-division multiplex line-finder.

The upper part of Fig. 12 will be recognized as being the same as Fig. 6(b), except that the variable-time pulse-trains are on the N -side and the fixed-time pulse-trains are on the M -side. An N -side circuit which calls over its forward signalling path is connected through a register-connector RC to a register-sender. When sufficient called-number signals have been received into the register-sender for setting up to commence, and when the marker is free, the register-sender is connected to the marker by the marker-connector MC . The called-number signals are then transferred at high speed into the group marker GM , which marks the appropriate group of circuit-marker leads CM . The circuit-marker leads are terminated on a pulse-marker multiplex

MMX similar to and synchronized with the connector-switch multiplexes MX1 and MX2. The marker multiplex delivers to a pulse-suppression gate circuit, PSG, a pulse train synchronous with the pulse train of every *M*-circuit which is marked on its circuit-marker lead. Hence, when the group marker is operated, marker-pulse trains for all the circuits in the marked group are transmitted to the gate circuit PSG, which is open unless inhibited, as described later, and thence to the marking-pulse highway MPH. Each of the channel-pulse generators CPG of the *N*-side circuits has a starting-pulse gate circuit STG commoned to the marking-pulse highway, but is normally closed. At the same time that it operates the pulse-marker multiplex MMX, the marker also opens the gate STG of the circuit being connected, by a signal sent through the register-sender, and allows the marking pulses to flow to the *N*-circuit channel-pulse generator CPG, which is started by the first pulse it receives and thereafter generates a train of pulses coincident with the marker-pulse train which caused it to operate. The channel-pulse generator therefore acts as the free-circuit selector; it is so arranged that it not only operates the *N*-circuit modulators to connect to one of the *M*-circuits, but also applies an inhibiting pulse-train to the pulse-suppression gate circuit, PSG, to prevent that *M*-circuit from being selected for any other connection so long as it remains connected to the *N*-circuit.

In the line-finder shown in Fig. 13, the speech and line-signal multiplex MX1 is also the marker multiplex. A calling signal from an *N*-side circuit starts a channel pulse through the multiplex MX1, to the highway H1 to which is connected a pulse-suppression gate PSG. Pulses which pass the PSG gate to the calling-pulse highway CPH are applied to starting-pulse gate circuits STG, of which there is one for each *M*-side circuit, and also, after conversion into a continuous signal by a pulse lengthener PL, as a group-marking signal to a free-circuit marker FCM which selects and marks a free *M*-side circuit. The marking is applied to the STG circuit of the selected circuit, and opens the gate for the pulses on the calling-pulse highway to pass to the channel-pulse generator CPG. As in the case of the selector shown in Fig. 12, the channel-pulse generator responds to a pulse and thereafter generates a train of pulses coincident with the marker pulse train which caused it to operate; the channel-pulse generator pulses also make the connection required, mark the *N*-side circuit as engaged by closing the PSG gate to the *N*-side circuit channel pulses, and signal over a test lead T that the *M*-circuit is engaged.

The descriptions given of multiplex connector switches with multiplex selection are of necessity only in outline. The point important to the paper is that the apparatus needed for an automatic exchange using these switches is little more than the speech and signal multiplex-terminals already described for the lines and trunks, and the channel-pulse generators. The register-senders are shared and are therefore relatively few in number.

(5) REGISTER-TRANSLATOR-SENDERS

It is well known that, for the control of a large network of many exchanges, the information communicated by the callers to register-senders in the form of called-number signals may have to be translated into other information before it can be sent into the automatic switches. Fully electronic register-translator-senders have been described elsewhere.^{9, 12}

(6) POST-SETTING-UP CONTROL

After a call has been set up through the connector switches, other controls have to be exercised, such as ringing the called subscriber's bell, tripping the ringing when the called subscriber

answers and operating the calling subscriber's meter. These controls are most economically vested in control units associated with suitable trunks between ranks of switches. Fully electronic control units present no difficulties; an example providing ringing and ringing trip is shown in Fig. 14.

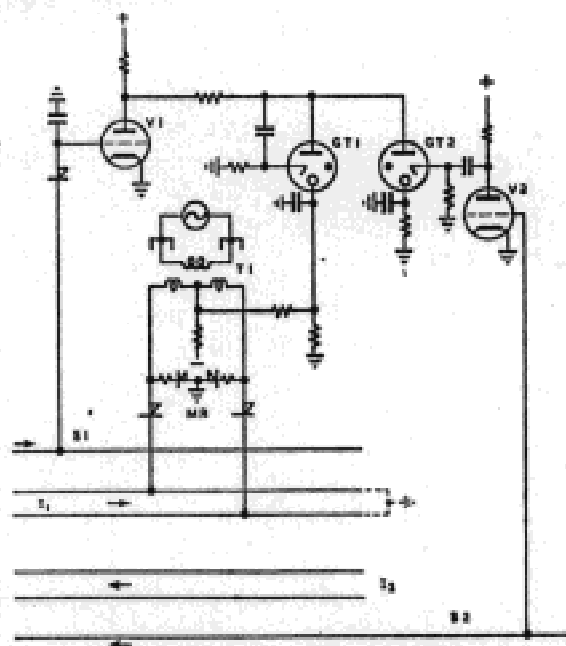


Fig. 14.—Ringing control unit.

In Fig. 14, a 4-wire speech circuit forming part of a trunk is shown by channels I1 and I2, with S1 and S2 the forward and backward signalling circuits respectively. When the call is set up, the forward holding signal, which is a change from earth to a negative potential, appears on S1 and cuts off valve V1. The rectifier and capacitor in the grid circuit of V1 endow that circuit with quick-operation and slow-release properties. The anode circuit of valve V1 controls the anode supply to a pair of cold-cathode gas-discharge tubes GT1 and GT2. GT1 is fired when valve V1 is cut off and applies to the forward speech channel I1 an interrupted ringing signal from a common source through transformer T1 and the rectifier switch MR. The ringing signal is a low-power a.f. one suitable for operating the line equipment shown in Fig. 7. When the called subscriber answers, the line signal on S2 cuts off valve V2, the change of anode potential of which fires tube GT2. This extinguishes GT1 and thus trips the ringing.

(7) POSSIBILITIES AND CHARACTERISTICS OF ELECTRONIC EXCHANGES

The previous Sections have been devoted to demonstrating that there are a number of practical solutions to the technical problem of the fully electronic automatic exchange. Necessities of space have limited to the barest outlines the description given of the shape of such exchanges; nevertheless the form in which an electronic exchange might be constructed is sufficiently apparent for discussion to be undertaken of its possibilities and characteristics.

It is perhaps opportune here to define what is meant by an electronic exchange. Parts of existing electro-mechanical systems may be replaced by electronic devices and units, but this would not make them into electronic exchanges. On the other hand, exchanges which are otherwise completely electronic

are very likely to have mechanical subscribers' meters and to be controlled by mechanical dials; they are not therefore completely electronic but are certainly intended to be covered by the term "electronic." Any definition of the term is therefore bound to be arbitrary to some degree, and in this spirit it is suggested as a practical definition that an exchange is electronic if the connector switches are electronic.

The philosophy of the paper is that any automatic exchange can be divided into parts, namely connector switches, selection system, register-senders and register-translator-senders, and post-setting-up control units. This division is certainly applicable to existing systems. It might be thought possible to produce some electronic structure so novel as to invalidate analysis based on known systems, but no such possibility is known to the author.

(7.1) Effect of Electronic Exchanges on Transmission Networks

A complete telephone system is a complicated network of transmission lines interconnected by the exchanges in innumerable combinations as demanded by the users of the system. The problems of connecting lines together by exchange switches with metal contacts having substantially no effect on the transmission are well understood. They include the effects of overall loss and the variation of overall loss of the lines with time, reflections due to mismatch of lines connected together, attenuation and non-linear distortion. The outcome is a practice of line construction and maintenance which provides a satisfactory standard of transmission at the minimum cost for the prevailing conditions. Electronic exchanges would make no difference to the fundamental problems or the principles of planning the system, but might compel or at least encourage changes in practice.

The transmission through electronic exchange-switches introduces extra possibilities of overall loss and variation of overall loss, of reflections and distortions. To that extent the problem is made more difficult and its solution in consequence less satisfactory. There are, however, more than compensating advantages to be gained. Connector switches which include one-way amplifiers in their transmission paths must provide 4-wire transmission through the switches; the amplifier gains will usually be such as to make the losses through the switches zero. There will, however, be a variation of the overall loss which has to be taken into account, but, in general, this will be more than offset by the greater gain which can be employed in line repeaters because of the 4-wire switching. In addition, a subscriber's line connected to 4-wire switches including amplifiers in the 4-wire channels is presented with a termination at the exchange which does not vary with the connections made. Side-tone balance at the instruments may then be improved, with a corresponding improvement in transmission.

Network planning tends at present to avoid amplifiers in local junctions at the expense of copper in the transmission circuits. In electronic exchanges with amplifiers available as a matter of course the tendency should be reversed, with a resultant saving in the cost of junction plant. For those systems which provide 4-wire switching in all exchanges, the greater gain thus possible in the line amplifiers should result in still further savings in junction and particularly in exchange line-plant. Finally, frequency-division and time-division exchange switches offer opportunities for carrying these types of transmission into local and short-junction networks as parts of an integrated transmission and switching system. It is not difficult to visualize, for example, the present unattended automatic exchanges being replaced by multiplex terminals connected by coaxial cable to a larger exchange in which is located all the switching for an area

of considerable size. The paper can do no more than draw attention to the possibilities and emphasize that the economics of electronic exchanges is not merely the production of exchanges which cost no more than electro-mechanical exchanges.

(7.2) Electronic Exchange Costs

There is little in the way of similar apparatus in production to give a guide to the capital cost of electronic exchange equipment in bulk production. However, from several points of view, one might reasonably expect that equipment to be much less expensive than electro-mechanical equipment of any sort. The number of parts required is a mere fraction of that required by the Strowger system, which is as cheap as any other known electro-mechanical system. Mass-production of the parts is thereby facilitated, and the costs of storage, handling and stock provision are greatly reduced. Tooling is greatly reduced, as is also the need for maintaining stocks of tools for obsolescent designs. New designs are quickly built up from the few standard parts, so that development costs should be small. It is however impossible to hazard a guess as to the ultimate cost, as it is almost certain that future work will bring simplification of circuit techniques and components.

Capital costs are recovered over the life of an exchange. The character of the exchange apparatus involved in the two cases suggests that electronic exchanges may have a longer economical life than electro-mechanical types, resulting in a lower annual charge for any given initial expenditure.

No new system would be adopted for replacing existing systems unless the labour absorbed in running and maintaining the new exchanges was less than that of their present counterparts. In electronic exchanges there would be nothing corresponding to mechanical adjustments and very little to clean. Labour would thus be limited to fault finding and replacement of faulty parts, which should be largely limited to hot-cathode valves. The number of such valves in a 10 000-line exchange may eventually be as low as 20 000, and with this number failures would occur at an average rate of some 50 per day if they had an average life of one year each. The cost of replacement and the effect on service would in these circumstances be prohibitive. At least a five-year life is a necessity, and there is good reason to think that this and more can be achieved. It is too early for the author to assess the possible effects of transistor development on this and other questions. With a modern and largely automatic fault-finding system, maintenance should be largely a matter of fault analysis by a small highly-trained and centralized body of men, and fault clearing a matter of changing small units. For these reasons it is not expected that the labour required would be anything like as much as is expended in the present exchanges.

The power absorbed by electronic exchanges may be expected to be a constant load due to the continuous current drawn by hot-cathode valves, plus a smaller load proportional to the traffic carried. Hot-cathode valves should be most numerous in multiplex-type exchanges, where each subscriber and junction line has one valve. At an estimated power of three watts per line, 25 kWh per annum is consumed for each line. The cost of this is only a small fraction of the total cost of telephone exchange service. A similar number of hot-cathode valves in the rest of a 10 000-line exchange would involve a total of about 60 kW of continuous load plus an additional load proportional to the traffic. The importance of this expenditure of power is probably not so much its cost as how to lose it as heat. However, any estimate regarding power requirements of electronic exchanges may very possibly be invalidated by valve and other developments in the next few years. The ultimate requirements are regarded as indeterminate.

(7.3) Service given by an Electronic System

The term "service" includes facilities, transmission and reliability as experienced by the subscriber. So far as facilities are concerned, there is no reason to think that there is any fundamental difference between electronic and electro-mechanical switching. Any facility which can be provided by one type of system could also be provided by the other. Any advantage which electronic switching might have in this respect would result from making a clean start with a new system.

The speed of operation of electronic switches should result in local calls being connected immediately the final digit is signalled by the subscriber. On the longest national call switched through exchanges all of which were electronic, the delay following the signalling of the final digit would be mostly determined by the rate at which number information could be passed on from exchange to exchange, and might reach three seconds. The quicker connection times can be regarded as a service improvement, although modern designs of electro-mechanical systems could achieve nearly as much.

If transmission is measured in terms of linearity and variation of gain through the switches, mechanical contacts, so long as they make good contact, are always better for transmission purposes than electronic switches. In electronic exchanges, which provide 4-wire transmission, even in the local exchanges, better overall transmission can be given than is provided by the existing system, despite the variation of loss through the exchanges. Connections made through existing types of exchange are occasionally subject to noise which causes a good deal of annoyance to subscribers. The cause of the noise is most often unsoldered joints, either intentional—e.g. wiper and bank contacts and relay contacts—or unintentional—i.e. dry joints. In electronic exchanges there need be no intentionally unsoldered joints; plugs and jacks should be avoided, except perhaps for power connections where they are not likely to do any harm; valves should also be soldered in to avoid the troublesome pin-and-holder unsoldered joint. The problem of dry joints is being successfully solved.¹³ Noisy connections through electronic exchanges should thus be rare, but electronic exchanges may have as yet unsuspected or so far unexperienced sources of noise. Poor insulation between heater and cathode of a valve could produce an unpleasant level of noise without giving a detectable fault. Hence, only experience can finally decide the question.

At present, reliability is perhaps the most controversial aspect of the subject of electronic exchanges. Whether this type of exchange can be made inherently more reliable than the best electro-mechanical exchanges is a question which only time will determine with certainty, but there are reasons for thinking that it will be so.

All systems are designed and manufactured to limits which if satisfied will guarantee reliable service. Defective service is largely related to the opportunities for the apparatus not to be within design limits and the extent to which inspection and test can discover the defects. With electronic apparatus each component once made is thereafter incapable of being adjusted. The number and character of the components are such that every one can be measured for its essential characteristics as precisely as may be required. These characteristics are electrical quantities directly related to the performance of the part in the system, and they are not affected by assembly, wiring, shipping and installation on site. With electro-mechanical equipment the conditions are not so favourable. The characteristics of mechanical parts cannot be so accurately specified or measured in terms of performance in the system as those of electronic apparatus. The parts are so numerous that sampling is more often used than the testing of every individual item. Many hand

adjustments have to be made, the quality of which depends on the skill of the adjuster and the vigilance of the man who makes the subsequent inspection. There remains the application of functional and other tests to detect faulty apparatus, but the efficiency of such tests is limited. The relay spring which has too little lift to ensure good contact is not readily detected by any externally applied test. Similarly, the pivot or hinge which is too tight or too loose is not easily detected until it gives trouble; what is worse, it may interfere with the proper functioning of the equipment for a long time until the trouble becomes apparent enough for detection. It is not suggested that electro-mechanical exchanges cannot be manufactured and maintained to give good service, but it is suggested that those conditions which can account for the small proportion of defective calls which occur on such systems may be largely absent in electronic exchanges. It may prove that electronic apparatus has peculiar sources of unreliability. What are sometimes referred to as catastrophic failures, e.g. valve-heater disconnections in electronic apparatus and loose and broken parts in mechanical apparatus, are at present more likely in electronic than in electro-mechanical apparatus, and the deterioration of rectifiers and other electronic apparatus over periods as long as 30 or 40 years is still to some extent an unknown quantity. Experience will show where the weaknesses lie, but at this early stage there is every reason to think that they can eventually be overcome.

(8) CONCLUSION

It is hoped that sufficient outline has been given of some possible forms of electronic automatic telephone exchange to convince engineers that it is only a matter of time before such exchanges will be produced in a form suitable for service in any size of network. Whether electronic exchanges will have, relative to the best electro-mechanical exchanges which can be produced, some or all of the attributes of greater reliability and less capital and maintenance costs is not so certain, but there is encouraging enough evidence in these directions to justify proceeding with the development of such exchanges. The consequential possibilities in transmission network design and construction are such as to justify electronic exchange development even if the other signs were not so propitious.

(9) ACKNOWLEDGMENTS

The author is indebted to numerous colleagues for assistance in the subject-matter and preparation of the paper. Acknowledgment is also made to the Engineer-in-Chief of the General Post Office for permission to make use of the information contained in the paper.

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DISCUSSION BEFORE AN EXTRA MEETING OF THE INSTITUTION, 13TH MARCH, 1952

Mr. R. A. Moir: This year is the diamond jubilee of the opening of an office called La Porte, Indiana, where Strowger's 5-wire 2-motion switch was first installed. Furthermore, it is 33 years since Grinted and Laidlaw read the paper* which they called "A look-see into the future," and more than a quarter of a century since Col. Sir Thomas Purves gave us his résumé of the logic and reasoning which lay behind the Post Office decision to use automatic exchanges.† It is therefore fitting that at this stage, when we are having a "look-see" into a further piece of future development, this paper should be read at a meeting of The Institution and be given to us by a senior officer of the Post Office.

The author has brought out the need to approach this subject without using any analogy with the existing equipment and apparatus. I have found, particularly in the education of young engineers, the fallacy of letting them get used to the apparatus from the line switch to the selector and connector switch, rather than educating them in the proper functions of hunting, selecting, storage, counting, etc., and then applying to those functions the tools and circuits, whether electro-mechanical or electronic, which are on hand for the purpose.

Another point which the author has brought out by his conservative treatment of the subject is that the problems of electronic working should not be rushed. I regard that as essential. We have to remember that the telephone relay has been used for many years in places quite remote from this field. That applies also to the single-motion switch to some extent. The 2-motion switch or other heavy mechanism is more or less confined to the telephone field. We are now faced with the possible employment of tools, components and circuits which may be used outside the borders of the telephone field, and therefore in my opinion we should proceed slowly, because we have to carry with us and consult with these other users on questions of coding, standardization, and other matters.

The next point which appeals to me is the necessity to take a very broad view of this question. I think that it is true to say that in 1924 the Post Office and other administrations were forced to adopt some form of automatic switching because it was becoming quite impossible to carry on, particularly in the large metropolitan areas, by the old methods. To-day, however, there is not the same pressure to adopt any further particular form of development, and therefore I think that we should benefit by gathering experience from outside the telephone field altogether.

Another related point is that, whatever the end product of this development, that product has to be sold. For very many years the Commonwealth and some other countries have, in the main, followed British Post Office practices, but it is difficult to forecast what they will do in the future. On this occasion, when we have the opportunity and the time to do it, we should try to bring this development into being on a broader basis and not rush into

its use for one particular purpose or for one particular country. We have an opportunity to see that what is ultimately produced is suitable in every way, and in that I include seeing that the development is such that the performance of what is produced is suitable for tropical and other fields.

As a member of the industry, I note with interest the way in which the author has skirted round the economic side of his story. Every author has to put in a paragraph about economics, and very few avoid getting bogged down at an early stage. From the industrial point of view it is essential that no false ideas on economics should become embedded in the Post Office contracts department.

The paper deals with full electronic switching, and I do not propose to raise a discussion outside that field, but I shall not be surprised if someone in some circuit laboratory in some years' time, or earlier, calls for his assistant and says "Look! I have made a great discovery. By putting in two uniselectors here I have saved a thousand tubes!"

Brigadier L. H. Harris: Forty-four years after Sir Thomas Purves reported that "assuredly more would be heard of automatic telephony," we find ourselves looking at electronic switching in much the same way, but with a much wider knowledge of the overall requirements of a telephone system and also much less isolated from other fields of electrical development in regard to common components and techniques.

At that time, in America, telephone engineers were waiting to see whether the public would mind dialling their own calls after the novelty had worn off. Now we know that the public are able and willing to dial not only their own local calls but, if necessary, trunk calls on a national basis.

If in those early days it had been foreseen that, owing to the demands of labour for entirely new industries, it would be not merely a question of marginal saving on operating costs but ultimately the more fundamental one of not being able to afford the man-power to expand fully the local services on a manual basis, a great deal of breath would have been saved.

From Figs. A and B, it will be seen that in 1921 we had in the United Kingdom about 16 000 operators handling 900 000 telephones and 840 million calls per annum, of which 57 million were trunk calls. Now we have 52 000 operators handling about 5·4 million telephones, of which 30% are still manual, the total calls approaching 4 000 million per annum of which 260 million are trunk (timed) calls. These figures for operators include exchange supervisors and a very large proportion engaged on auxiliary exchange work, sick, holiday, etc., reliefs and P.B.X. staffing. The changes in conditions of staffing and methods of operating during these 30 years make it difficult to estimate precisely the operator savings due to the introduction of automatic working for local calls, but my own estimate is 18 000 to date. What is not difficult to see is the relatively small number of trunk calls dealt with per member of the exchange staff per annum and the vital necessity for curbing the increase in numbers as the system and the trunk service continues to expand—and

* "Telephone Service of Large Cities," *Journal I.E.E.*, 1919, 578, p. 158.

† "The Post Office and Automatic Telephones," *ibid.*, 1925, 63, p. 617.

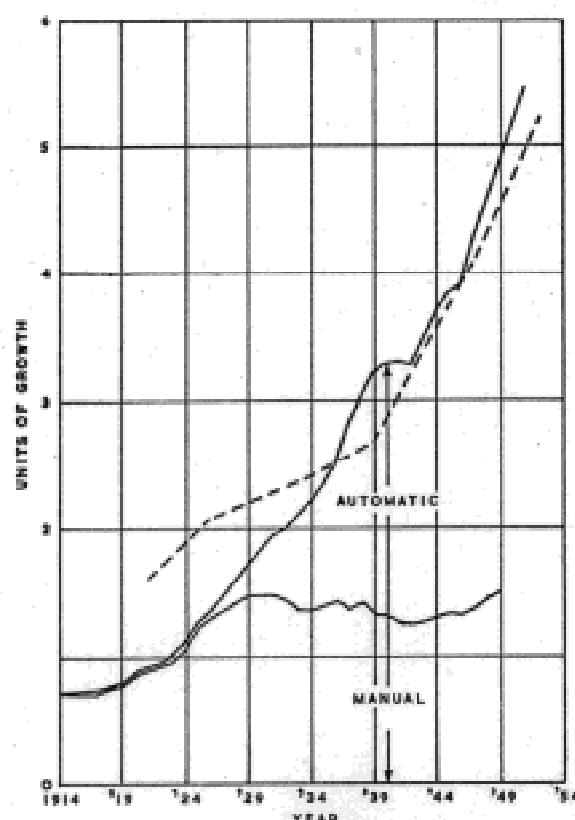


Fig. A.—Growth of telephone stations and staff.

— Stations; units are multiples of 10^4 .
 ---- Operating staff; units are multiples of 10^4 .

It is clear that it has a long way to go yet. It seems certain that the brain-like possibilities of electronic register-translators are of great potential value in increasing the percentage of calls to be completed without the intervention of an operator and in a more efficient and flexible manner than would be possible by electromagnetic methods.

In its early stages a newly developed non-mechanical system would be competing in reliability and costs with a successful and mature mechanical system; nevertheless, it would have to show promise of reduced annual costs and of putting a brake on the increasing numbers of maintenance staff. It must be remembered, however, that only about 20% of our 60 000 technicians (20 000 in 1921) are concerned with the nature of our switching equipment and of this 20% only differential savings would be possible.

The author visualizes the economic dispersal of multiplex equipment into the exchange areas; the increasing use of wide-band cables for various purposes and the need for conservation of copper would seem to encourage an early blessing of this idea. Even so, any new system would have to work on the existing line plant, and full benefits would come later.

Regarding accommodation, I hope that the line of development will keep open the possibility of some form of unit construction or transportability, say, up to 2 000 lines, and that special buildings will not be essential. We have had many difficulties in the past which could have been overcome more readily with equipment designed for flexibility and transportation.

Finally, while we are trying to advance the telephone art from our present very good mechanical system, I hope that we shall not at the same time be introducing even more complicated

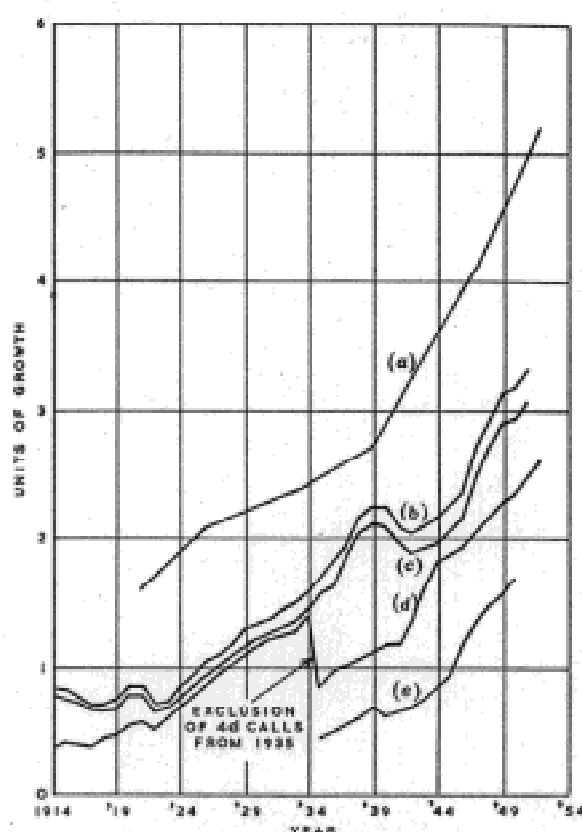


Fig. B.—Growth of local and trunk traffic, operating staff and trunk circuits.

(a) Operating staff; units are multiples of 10^4 .
 (b) Total calls; units are multiples of 10^6 .
 (c) Local calls; units are multiples of 10^6 .
 (d) Trunk (inland) calls; units are multiples of 10^6 .
 (e) Trunk circuits; units are multiples of 10^4 .

mechanical devices in connection with the detailed billing of trunk calls. With the prospect of a continuing need for the conservation of operating, maintenance and clerical force as still more demands are made on productive man-power and, bearing in mind the reducing proportion of distant dependent costs, I believe that the only lasting solution to the charging and billing problem will be found in some form of bulk metering applicable eventually to the large majority of subscribers.

Dr. J. H. Mitchell: I think it should be stressed that the developments recorded in the paper are, of necessity, still very much in the research stage. I think the author perhaps understates the possibilities of the gas-discharge speech tube. In the work for which I have been responsible in this field, cold-cathode tubes having an attenuation of about 1 db and a noise level better than -70 db at 1 mW have been used, and a perfectly satisfactory response for commercial speech has been obtained. Further developments are practicable and are being investigated. Apart from the possibility of using such tubes extensively throughout the speech path of an exchange, I suggest that even with multiplex systems there may well be an application to the first stage providing quite simple switching.

In the paper the subscriber's apparatus appears rather costly—at a quick estimate, probably twice that in the existing type of exchange—and as this apparatus is repeated for every subscriber, either this would need to be cheapened or saving will have to be obtained in the remainder of the exchange. One point not discussed is the wiring, which in mechanical exchanges is some

10% of the total cost, but in an electronic exchange the impedance will be much higher, and thus the liability for crosstalk much higher, demanding greater screening. Is it likely that there may be an appreciable increase in the costs and difficulties of wiring an all-electronic exchange?

I think that it is unfair to make a detailed comparison between the number of parts in an electronic exchange and in the mechanical system. The circuit engineer is apt to take a valve as a component and forget the intrinsic tooling that goes to make a valve and the large number of detailed components there are inside it. The troubles involved in such things as welds and the high vacuum necessary for long-life valves are sometimes overlooked. When all these things are added up and consideration given to the tooling and production problems of the various valves required for an exchange, it may well be found that this item is approaching in its costs and complexity that of a mechanical switch.

Another point is that a telephone exchange lasts about 25 years. Extremely long-life components are therefore required, and this demands long ageing and testing schedules and very careful process control. Unfortunately long-life components are apt to be very expensive, even when mass produced. If an electronic exchange is to be produced then, bearing in mind this very long life, an engineering performance will be required which is higher than has been customary in the best-quality radio or communication equipment that has yet been produced. Once the problem has left the research field there will be a severe problem for the engineer and also subsequently in manufacture, because the standards of inspection and testing will have to be extremely high.

I find it difficult to follow the trunking of the time-division multiplex selector. Whilst the author gives some indication in Figs. 12 and 13, I feel the ideas are still very incomplete. The author showed a slide and made reference to a more detailed arrangement. Could he please elaborate the general trunking arrangements, for therein lies the crux of the whole system?

With regard to the author's remarks on cost, there are several points that still require very careful thought and which are likely to prove extremely difficult in estimating anything like the true cost. I think at this stage it is anybody's guess which way the final costs will go.

Mr. Moir said that there is no great hurry in this development: in point of fact we must not hurry or else we shall find too late that the magnitude of the problem has not been fully appreciated, that the first costs and reliability are disproportionate, and a host of troubles will come home to roost.

Mr. W. Six (*Holland*): I do not agree with the author that cold-cathode tubes must always have an attenuation of at least 1 db. I have worked on electronic switching for some years, and I have a small electronic exchange working with cold-cathode tubes in the speech path, and there the attenuation is about 1 db per switching stage. For a small exchange that does not matter so much. I agree that for larger systems it would be too much, but I do not think that there is a limit at 1 db, because the tubes about which I am speaking have a current of 6 mA, and the attenuation is inversely proportional to the current in the normal region of the gas discharge. It is therefore easy to take a higher current, say 12 or 20 mA or even more, and have an attenuation of 0.5 or 0.3 db, or even less.

There is also another possibility. We have been able to make cold-cathode tubes with constant negative resistance from zero to 4 kc/s, and with this negative resistance we can have still smaller attenuation. I think we can therefore say that it is not correct to state that there is no possibility of having tubes with attenuation lower than 1 db.

Capt. J. Legg: The author concludes in Section 8 that it is

only a matter of time before such exchanges will be produced in a form suitable for service in any size of network. Before we can produce a range of electronic exchanges for different sizes and types of network and make it available for and attractive to operating concerns who are in the market for switching equipment, a tremendous amount of work remains to be done. The author has made it clear that many alternative means exist. All these alternative plans have to be further developed and examined, and if we are to make the best use of the restricted force of specialists available for work in this field, decisions will need to be taken, in the light of facts as they come forward, to proceed or not to proceed in this or that direction and perhaps to concentrate in one field to the exclusion of some others.

This type of need cannot be met by individual research or development groups or individual manufacturing organizations; nor can it be carried through by the Post Office alone. It needs the concerted effort of all these interests. I would therefore urge the necessity of getting together now and setting up machinery to provide for a concerted effort in all these fields, so that this very extensive and of necessity long-duration work shall not take unduly long and shall not make undue demands on the staff available.

The author has been careful to point out, when discussing the economic possibilities, that it is not merely a matter of the production of exchanges which cost no more than the equivalent electromagnetic exchanges, but that in fact we are led, in contemplating electronic exchanges, to contemplate also fairly radical changes in the rest of the network, notably on the transmission side, which may result in further savings in junction and in line-plant.

This is attractive as making progress possible in directions which are difficult or impossible with our present layout in Britain. We have to remember, however, that if the economics of the new exchanges require a money entry on the credit side on account of savings due to improvements in the network elsewhere, the process of changing this layout is of necessity very slow, and the financial benefits will be slow in being realized. This does not mean that we should ignore the long-term potentialities, but it does mean that, in weighing the pros and cons of electronic switching, a long-term study of probable network developments becomes part of the process and adds to the work involved in arriving at a reasoned judgment on the problem which we are considering, namely the advisability or otherwise of introducing electronic exchanges.

Mr. E. P. G. Wright: The basic question is whether electronic telephone exchanges can provide satisfactory service with less expenditure. Accommodation, pension, interest, depreciation and administrative charges will not easily be reduced, and it is therefore in the 22.5% chargeable to maintenance that savings must be found. Having established a greatly reduced maintenance force it will be necessary to ensure adequate supplies of "highly trained men" who may also be in demand for various industrial electronic installations.

The percentage reduction in expenditure must be substantial to attract overseas administrations having some modest annual extension programme. Any possibility that electronic exchanges might obviate the need to continue production of mechanical exchanges is important, because the simultaneous engineering and manufacture of different systems must result in increased cost. It must be recognized that reduced charges for junctions and exchange lines due to exchange amplifiers or time-division switching may involve the segregation of line plant in multi-exchange areas into separate groups whilst mechanical exchanges remain in operation (see Fig. C). Fig. C illustrates the importance of developing a compromise system, possibly not "all electronic," but designed to reduce maintenance by eliminating

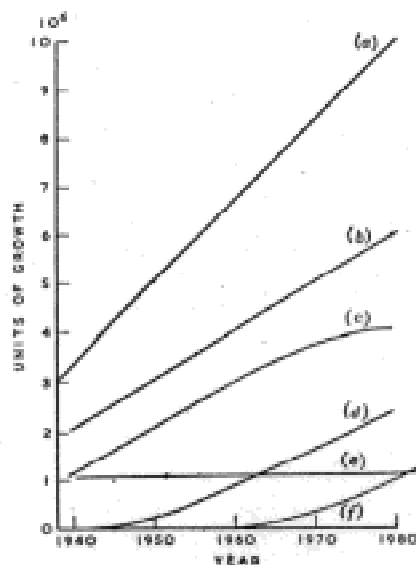


Fig. C.—Forecast of telephone expansion.

- (a) Total stations.
- (b) Total exchange lines.
- (c) Mechanically operated exchange lines.
- (d) Shared-service exchange lines.
- (e) Manually operated exchange lines.
- (f) Electronically operated exchange lines.

parts subject to wear and adjustment and designed for extending existing installations. Many of the advantages gained by the use of electronics can be applied to existing systems. Developments on these lines are to be expected from various research organizations.

There is no overwhelming reason for an immediate decision, and some study might advantageously be made of the possibilities of memory devices using cathode-ray tubes, mercury columns or, preferably, magnetic drums. Such devices might make selections and replace timing relays, message registers and traffic meters. A director or regenerator might take the form of 2-in $\times$ $\frac{1}{2}$ -in bar of nickel with a track circuit common to a number of storage circuits. The ease and speed with which such memories can be scanned provides an opportunity of reducing the clerical effort involved in the accounting of local and toll calls. A wide employment of electronic equipment to undertake such clerical operations in industry is likely, and in the telephone industry the experience necessary to design such systems is available.

Progress on electronic switching during the last 20 years has been disappointing in comparison with the growth of scientific knowledge. The development of an electronic exchange based on carrier technique, such as frequency-division multiplex, has made little progress. Large sums have been spent on electronic-switching development in several research establishments, and the models demonstrated in the past have clearly emphasized the need for caution. This suggests that it will be wise to gain wide experience before deciding on any new system which might become obsolete within a decade, to accept simple maintenance as a more important target than an all-electronic exchange and to recognize easy extension of existing installations as more valuable than potential improvements which cannot be fully realized so long as the majority of exchanges are mechanical.

Mr. F. O. Morrell: Judged from the paper, the development of the electronic exchange appears to be an unusual one. Major developments usually begin with a requirement or a new facility; in this case the problem appears to have been that of making an electronic exchange which does no more than the existing electro-mechanical exchange but in which, by the author's

definition, the connector switches are electronic. The result appears to be that undue attention is given to electronic connector switches. It is a simple matter to devise a connector switch using valves instead of contacts, but it would be very expensive; hence it is necessary to use a multi-channel carrier or time-division system as described in the paper to economize in the number of connectors installed. The maintenance of carrier systems, however, is very expensive and it is difficult to believe that it is likely to pay within a single exchange. The alternative is a metal-to-metal mechanically operated contact, and if this had only to complete speech paths it should not be an impossibly difficult switch to design and produce. In Section 2.2 the author states that metal-to-metal contacts produce little attenuation, attenuation distortion or non-linearity in transmission, which effects are always present in electronic switches. Such contacts would seem to be a more promising line of approach to less cost and better service in exchange production and maintenance. Admittedly, existing mechanical switches are not wholly satisfactory, but the functions of selectors and connectors are largely combined and the former have much more onerous work to perform. An electro-mechanical connector would be a very simple piece of apparatus. Used on call basis, it would have to operate comparatively few times, i.e. much less often than a switch in the control equipment, and the timing would be unimportant. Another means of approaching the ultimate electronic exchange would be to examine existing exchanges and ask how electronics can best be used to ease the more difficult problems of design, construction and maintenance, and if possible by this means to reduce the cost of, or improve the service given by, the exchange. Many instances where electronics might be of value come to mind; most of them are in the control equipment of the exchange. A recent paper* refers to the heavy demands made on relays in the No. 5 crossbar system, which must therefore have higher operating speeds, longer life and greater adjustment stability. Electronics might be used with advantage for such purposes.

Mr. D. A. Barron: It has been part of my duty to make a close study of electro-mechanical exchange-switching systems throughout the world. Although they vary considerably in principle, the best of them are able to give very good service, and improvements continue to be made. The fundamentals, however, have been so thoroughly explored that it seems unlikely that any development is now possible which could provide a new electro-mechanical system having outstanding advantages over anything existing. On the contrary, electronic exchange switching, in part or whole, is almost embryonic, and any present achievement may well appear rudimentary in a few years' time. I had therefore expected to hear very ambitious hopes expressed by electronic engineers during the discussion, and was prepared to sound a note of caution. In view, however, of their general pessimism, a word of encouragement from the development and maintenance side seems appropriate.

Primarily to obtain field experience, we have been introducing certain electronic apparatus into the present system, with promising results, and further applications of electronic techniques are projected. The author has referred to the electronic register-translator, and it may be of interest to mention that six such registers, with a common translator, have just been put into service, and have so far carried live traffic for a week without any fault. As evidence of the rate of progress, it can be stated that a newer design, showing a saving of as much as 60% in both size and power consumption, has already been worked out.

Our maintenance experience on electronic exchange equipment is necessarily limited, but there is reason to think that valve life

* SLAUSON, W. C.: "Improved U, UA and Y Type Relays," *Bell Laboratories Record*, 1951, 29, p. 466.

may not prove to be so predominant a factor as the author appears to suggest. In this connection, does he consider that electronic systems will be able readily to accommodate the variations in characteristics, with time, which occur in such components as resistors and capacitors?

I am disposed to discount forecasts of excessive power requirements for electronic systems, in the belief that developments will reduce the consumption to an amount which will not cause undue difficulties. The present use of battery capacity as a safeguard against mains failure would presumably, however, need to be superseded by the provision of standby prime-mover/generators. Regarding the mains supply, does the author feel that the sort of frequency and voltage variations which we now experience would be a source of difficulty in full electronic working?

Although an electronic switching system would avoid certain contacting difficulties inherent in electro-mechanical systems, it might contain as many, or even more, soldered connections, and dry joints could cause serious trouble. In this connection, has the author any views as to the possible advantages of printed circuits, or spot welding as an alternative to soldering?

Mr. G. C. Hartley: As several speakers have emphasized, switching development now is much more complicated than it was 30 or 40 years ago. To-day, any new techniques have to take their place beside millions of pounds' worth of existing plant and to work into an integrated network which is as vast in its extent as anything that any branch of engineering can produce. That means, in my view, that the supreme requirement of any new techniques must be versatility and flexibility. I do not say for one moment that I believe that the electronic techniques cannot and will not attain the necessary standards, but I believe that very close scrutiny is required.

For all its faults, one of the virtues of the old step-by-step system has been that so much could be done with one relay, a uniselector and a 2-motion switch, and that throughout all its operations there has been one uniform mode of inter-stage signalling. Modern trends, such as subscriber trunk dialling and so on, have made this position no longer tenable, but we have to be cautious about rejecting it completely.

The author claims that electronic circuits are flexible and that the components are simple and few. I think that there is a slight danger that he is confusing components with tools. The tools which make up a system such as this are assemblies of components and combinations, which in many cases are quite complex and require a good deal of investigation and testing, and I do not feel certain that the conclusion can be drawn that these are inevitably as simple to handle as the relay and uniselector.

Time multiplex is a fascinating system, but many things need to be established. For instance, is amplitude modulation ever going to provide the necessary crosstalk, signal/noise ratio and other characteristics needed? If amplitude modulation is not going to do so, or in the process of doing so will be too expensive, we come back to the point which Dr. Mitchell mentioned, that perhaps the pulse multiplex could be more effectively used for the heart of the system, and the fringes, the initial and terminal switches, could use some other technique.

With regard to direct audio switching, I believe the author is somewhat pessimistic and I am certain that transmission loss and noise can be improved enormously.

I agree with the author that electro-mechanical devices must inevitably remain in the ultimate system in some degree. I think that electromagnetic relays perform functions which cannot be entirely replaced by other devices, although to be in keeping with any electronic technique the relay must take a very different form from the one we know now; but why, to illustrate this point, does the author refer to the subscriber's message register?

I should have thought that that was the first and not the last to be replaced.

Dr. J. R. Tillman: The paper indicates that the number of switches through which a call must pass may be about the same in electronic systems as in electro-mechanical systems. Clearly the distortions permitted per switch must be small if a good standard of transmission is to be achieved, and it is essential to know that these will not be exceeded with simple designs.

The development of carrier telephony has long provided us with most of the data required to assess the transmission through a frequency-division multiplex switch, enabling us to specify the quality of modulators, filters and amplifiers required. On the other hand, many of the basic data for time-division-multiplex switches have had to be gathered since 1947. Only amplitude modulation has been seriously considered; it requires much simpler circuits than do other modulators, e.g. width and pulse-coded, although these have been lightly examined. Incidentally, the disadvantage of amplitude modulation which the radio engineer recognizes does not apply here.

Data can be given in some detail for the switch described by the author, which has 100 channels connected to its fixed-pulse-time side, a p.r.f. of 10 kc/s and a pulse duration of 0.3 microsec (corresponding to a channel duty ratio of 3:1 000), after its basic circuit (Fig. 6) has been converted into a practical circuit with all components small and working well within their ratings, power supplies of the quality used in telephony, and supplies of pulses of near-rectangular shape, constant amplitude and time-relativities constant to within 0.03 microsec. The great importance of the pulse lengthener (MR5, C5, R5) which makes amplitude modulation so attractive in practice ought first to be noted. It can give a gain within a few decibels of the high insertion loss (≈ 50 db) which accompanies the low duty ratio—which would otherwise have to be provided by conventional amplification—and makes possible a satisfactory signal/noise ratio without special precautions. It does, however, introduce some attenuation distortion, change of gain with time and a little non-linearity.

The transmission properties are then as follows. The noise produced from all sources does not exceed a few tenths of a millivolt across the subscriber's line for an average call. Non-linear distortion is not excessive. The modulation is transmitted through germanium diodes, but only when they are working in substantially linear regions. When the switch is handling the equivalent of 1 mW of signal from the subscriber it produces a second-harmonic content which is about 38 db below the fundamental. It is not easy to be definite about crosstalk yet, but despite an unexpected source from some germanium diodes* and despite the growth of interface impedance in valves used as cathode-followers, e.g. between H1 and MR6 in Fig. 6, satisfactory attenuations (i.e. >65 db) have been achieved. Some further study is needed, however.

Attenuation distortion is introduced by the transformers, the filter, the capacitor C1 and, more especially, the pulse lengthener. We have shown that the formula Kleene† developed for idealized lengtheners applies fairly well to simple practical forms. A distortion of 0.5 db is introduced at 2 kc/s and of 2.5 db at 4 kc/s. A simple equalizer will largely compensate for this.

The constancy of gain of the switch has so far been marred mainly by ageing of the thermionic diode used as MR5, which is necessarily operated under suppressed-emission conditions for at least 99 out of each 100 microsec. There is a great need for a miniature long-life diode. When changes of supply voltage, ageing of germanium diodes and tolerances on components are

* TILLMAN, J. R., and YENGE, H.: "A Note on the Decay of Current in Germanium Diodes," *Philosophical Magazine*, 1950, 41, p. 1281.

† KLEENE, S. C.: "Analysis of Lengthening of Modulated Repetitive Pulses," *Proceedings of the Institute of Radio Engineers*, 1947, 35, p. 1048.

also taken into account, the standard deviation of the loss through a switch may be 0.75 db. A call between a typical London and a typical Birmingham subscriber might be expected to have a standard deviation of transmission loss of 4 db.

Design is eased and distortions much reduced if the number of channels connected to a switch is reduced to below 100. Thus, if only 70 channels are connected, crosstalk attenuation is better by perhaps 10 db, and the figure for gain variation is halved. More switches are, however, needed in an exchange; can the author give any idea of the relationship between overall cost and switch capacity?

Mr. G. H. Foot: By the time electronic devices such as we are considering come into use the telephone network of each country will be planned as a single automatic system for the whole country. It would thus be advisable to approach the subject by examining the overall requirements of such a system, and debate the advantages of electronic telephone systems rather than those of electronic telephone exchanges.

A telephone system is not "a complicated network of transmission lines interconnected . . . as demanded by the users of the system." A user of the system demands only that two particular telephones be interconnected. The most useful concept is that each cell carries a label with the address of its destination. The label is scanned and a circuit chosen so that the call travels by the shortest route. A direct circuit will be selected wherever possible, but if there is congestion the call will be switched to other routes and make its journey in a roundabout manner. The direct routes require no switching; switching must be avoided wherever possible.

When a connection has been established, however, why must it be broken at the end of the conversation? It may be needed again for most of its distance. Why cannot calls which are going the same way be collected together and switched together? Bundles of twelve would be convenient. At any particular moment more than one-half of the circuit paths engaged are not being used. Normally only one party speaks at a time, and the return paths are idle. Why cannot paths be allocated as required?

These are some of the things which seem possible with an electronic system, if existing mechanical systems are not taken as a model. I have in mind a system which places a permanent label on each call. Every outgoing circuit carries the labels of all places to which it has access. Apparatus is used to examine the label of the incoming call, look for a circuit with the same label and make the appropriate interconnection. No dialling, no registers, no translation and no by-path are required. Electronic methods are necessary, however, to examine the labels at the required speed.

The author proposes a set of modulators in the switch room which will perform essentially the same function as those in the transmission room. It may be that the greatest attraction and the economic justification of an electronic system will prove to be its ability to merge switching and transmission elements to provide one system with both functions. With this approach the viewpoint is changed, and frequency rather than time multiplex may be favoured. The new carrier systems with a wider frequency separation, now making their appearance for other reasons, are a development which is significant in this connection.

A large variety of new features are practicable with electronic methods. Non-decimal systems can be used, e.g. to a base of 100. A base of 25 can be realized with an alphabetical system, and if that is done every telephone in Great Britain can be defined with a 3-letter/4-figure code. The most radical changes however may follow the emergence of new units from the laboratories, and at the present stage it may well be that the system of the future is still in the hands of those equipped to carry out fundamental research.

Mr. R. W. Palmer: If we are going to have an electronic system of any of the kinds referred to, it seems that we have to "sell" such a system on the score of costs, which includes maintenance facilities, and really good service; but that is the one point on which the author appears to be most uncertain, and he refers, perhaps rather vaguely, to the view that "experience will show" and "time will tell." Whilst he refers at some length to the fault liability of components, he says very little about the other aspects of maintenance which are peculiar to telephone exchange systems.

The maintenance of a telephone exchange is complex and must be linked closely with the design of the system. To use a simple analogy, the oil lamp has a very low fault liability; the light very rarely goes out if we give it daily attention, cleaning it and seeing that it has enough oil. But that does not mean that electric lighting is not better, in spite of the fact that it has a higher catastrophic fault liability. We should not dream of spending a lot of money on giving it daily attention, fiddling with the lamps and testing them to see if the light still comes on. Similarly the use of testers as for step-by-step systems may not be the best maintenance procedure for electronic or other common-control systems. The author has already pointed out that electronic systems require a new fundamental approach to the trunking design, but so also must there be a fundamentally new approach to the maintenance problems.

This, therefore, is an appeal to system designers to stop treating maintenance and service as something on which to wait and see what the other fellow does about it. Maintenance should be positive and an essential part of the design. The electronic system should provide a faith that the inherent fault liability will be so low as to make each fault a rarity and something justifying an automatic alarm and automatic tracing facilities. In this way we will not only produce a good electronic telephone switching system but will be able to "sell" it with confidence.

Sir Archibald Gill: This development started when I was Engineer-in-Chief of the Post Office, or perhaps even before that, and was continued during my period of office. This is an historic occasion; it is the first time, I believe, that an electronic telephone-switching equipment has been shown openly at a meeting in Britain. I hope that this work will be proceeded with and that further development will take place as actively as possible. In the past there have been many inventions and discoveries which originated in Britain which have been taken up abroad and brought to fruition and sold back to us at a very high price. I hope that in this case we will go ahead and develop this electronic exchange.

Some reference has been made to economics, but it is ridiculous to talk about economics at this stage and to try to compare the performance of this system in its embryonic stage with that of electro-mechanical systems now approaching the limit of development after 63 years' use.

It is understandable that manufacturers are anxious that the impression should not get abroad that a fully developed electronic system is available for installation, because it will be some years before this is so. On the other hand, it is important that it be known that work of the kind described in the paper is going ahead in this country and that Britain is likely to be able to offer a satisfactory electronic system as soon as or before other countries.

When the automatic telephone was first brought to Britain from the United States the Post Office very wisely provided an installation at the G.P.O. Headquarters, so that the engineers could acquire first-hand knowledge of the working and maintenance of the system. I think it would be well to repeat this procedure and to install, say, a 1 000-line electronic exchange as

quickly as possible, in order to get some real information of its behaviour under actual working conditions.

Mr. A. Davison (*communicated*): The author appears to take the view that only the most complete form of common control, i.e. systems such as time-division for the speech path, could be used for electronic exchanges. Whilst it would not appear economic to use the equivalent of the step-by-step system, an intermediate stage such as systems using individual speech-path circuits with common control for setting up the call cannot be ignored. Failure of any common-control elements would in this case affect at the most only the setting up of calls, whereas in a time-division system a simple failure could affect a large part of the exchange—in particular, subscribers actually talking to each other.

I feel that the author somewhat cursorily dismisses the use of cold-cathode tubes as speech elements. Noise in these devices can be made low enough to meet any telephone requirements, and it would appear that the hope of comparatively low insertion loss may yet be realized. The use of these tubes as multiple elements for large exchanges would probably be decided by economic factors, since no serious technical difficulties are apparent at this stage, i.e. in the form of control, crosstalk, etc. Life under such multiple conditions should be long, since the tubes would be used comparatively seldom.

A further advantage is that a half-way stage, using mechanical devices controlled by circuits basically similar to those used for the cold-cathode speech paths, could be introduced. This would give experience in the handling of electronic control, before the wider problem of replacing the speech-path contacts is attempted.

It is not apparent why the use of 4-wire trunking is so attractive, since undue complication of such equipment as the subscriber's line circuit would appear to be the direct result of such use. Does a system such as that described in the paper demand this trunking? The number of thermionic valves appears to be very great in the system described, including one per line circuit, or termination. Is this a result of the inherent high power-loss in the system and the low power which can be handled by the rectifier elements?

Finally, has the author applied these principles to units bigger than 100 lines and with more than one stage of selection?

Mr. J. McGavin (*communicated*): In Section 1 the author quotes erosion of contacts due to current passage as a defect of electro-mechanical devices from which electronic switches are free. However, it would appear that, in the course of time, cathode erosion will constitute a similar defect. The author seems to envisage this himself at the end of Section 7.3.

THE AUTHOR'S REPLY TO THE ABOVE DISCUSSION

Mr. T. H. Flowers (*in reply*): An important feature of the discussion was the wide support for the development of partially electronic exchanges before fully electronic ones. Partially electronic systems are not mentioned in the paper, chiefly because I am strongly against any considerable expenditure of effort at present on them. Changes can be brought about in already well-developed telephone systems only very slowly, as several speakers remarked, and the cost of introducing new apparatus absorbs some of the potential savings, often costing in the early stages more than is saved. To make a change in two stages increases the development and engineering effort required, delays the ultimate benefits by perhaps a whole generation and incurs the expense of changing twice over. The possibilities of fully electronic exchanges should therefore be known as soon as those of partially electronic ones, which at present requires the expendi-

The question of power supplies is dismissed rather lightly. Besides the difficulty of heat dissipation there will be a number of serious problems in connection with the distribution, fusing and supervision of the heater and anode supplies. The Post Office standard soldered-in valve may be adequate from a laboratory point of view, but it is unsatisfactory from both manufacturing and maintenance aspects. The author would be more accurate if he said the detection, rather than the problem, of dry joints is being successfully solved. It seems rather naive to say that the characteristics of electronic apparatus are unaffected by assembly, wiring, shipping and installation. At the frequencies the author is suggesting, alterations in layout can make a large difference to the stability of circuit performance. Frequencies used are certain to increase rather than decrease, thus placing further emphasis on the exact spatial relations of components and inter-connecting wires.

At several points the author suggests that reasons exist for thinking that the reliability of electronic exchanges will be considerably greater than that of electro-mechanical ones. It would have been helpful if some of these reasons could have been stated specifically. I agree with the author that initial weaknesses will eventually be overcome, but I do not think he emphasizes sufficiently the aspect of time.

Mr. G. W. Thompson (*communicated*): The author is inclined to the view that contact-making devices are prone to trouble and should be avoided at all costs. If, however, one examines the possible weaknesses in modern telephone exchanges of the electro-mechanical type one finds that, in general, contact-making devices are troublesome only if the component is operating in a heavily worked part of the system, e.g. common-control circuits, register-translators, etc. Furthermore, even with fully electronic schemes as envisaged by the author one must bear in mind the fact that the satisfactory operation of the whole system relies not only on the exchange apparatus but also on the service provided by the telephone instruments. In each instrument there are such contact-making devices as the dial and switch-hook springs, which are not heavily worked, receive very little maintenance attention and which, as many years of experience have shown, give excellent service.

I should like the author's confirmation that with either a frequency- or time-division switching system it is necessary, as indicated, to have a process of modulation and demodulation at each switching stage. If this is so, the process must be repeated four or five times in a 10 000-line exchange, which not only affects the economic aspect but may result in an overall loss in faithful reproduction of the original audio signal. These problems would not arise if individual electronic speech channels or contact-making devices were employed.

ture of more effort on the first than the second. Only if after a reasonable time the fully electronic exchange showed insufficient promise should the partial version be considered. It was brought out in the discussion that, operationally, there is no urgency in the question, and I suggest that five years would not be an unreasonable time to defer judgment on fully electronic systems. At the same time, the more plant of an existing kind which is installed, the less the advantage to be obtained from changing to a new type. For this reason research and development should be continued with all speed. Previous experience is not an infallible guide, but the proposal to change from manual to automatic exchanges produced very much the same situation as the present one and the same arguments as are now applied to the question of changing the type of automatic apparatus. Semi-automatic exchanges had a brief vogue and one cannot

help thinking that the same may be true of the semi-electronic system.

On the question of the best type of electronic exchange several speakers were plainly in favour of the gas-discharge tube for connector switches. Some interesting evidence was produced to show that the paper was pessimistic in its assessment of speech transmission through such tubes. Nevertheless, until it is shown that problems of signal transmission and very cheap tubes to offset the large numbers required have been solved, the case for gas-tubes for large exchanges is not, in my opinion, very strong. The paper admits their possibilities for very small exchanges.

I am in agreement with Mr. Moir's broad view of the electronic-exchange situation. The data given by Brigadier Harris and the remarks of Sir Archibald Gill are valuable contributions to the wider issues involved, and I strongly endorse their final points concerning, respectively, bulk metering and the urgency of practical experience with electronic exchanges.

In reply to Dr. Mitchell, I think it is wise not to look too closely at present at the cost of any particular part of an electronic exchange, but to be satisfied that it is not hopelessly expensive. If the switches provide 4-wire transmission there will be a proportional increase in the cabling, but otherwise I do not know of any reason why wiring should be appreciably more costly and difficult than in present exchanges.

The question of long-life electronic components is difficult, because one cannot be really sure that a component has a life of a certain number of years in less than that number of years. However, estimates can be made from performance over many fewer years, and on the whole I think the signs are encouraging.

Fig. D associates Figs. 12 and 13 of the paper in a simple exchange with one line-finder and one selector stage. Trunking arrangements with any number of line-finder and selector stages may be used in electronic exchanges in the same way as in

existing exchanges. I entirely agree with Capt. Legg's remarks, but Mr. Wright is more cautious in his approach to electronic exchanges than I should have expected from his prosecution of progress in other directions.

In reply to Mr. Morrell, I suggest that a frequency or pulse carrier system forming part of an electronic switch is sufficiently different and simpler than systems for long-distance transmission for the maintenance costs to be appreciably lower. I think he over-emphasizes the apparently unfavourable comparison in the paper between metal-to-metal contacts and electronic equivalents. If connections made electronically can never be as good as those normally made by metal-to-metal contacts, they can still be good enough and relied upon to operate properly every time, whereas the metal-to-metal contacts occasionally fail completely. Improvements in metal-to-metal contacting switches to produce near-certainty in the contact-making are expensive and provide one of the reasons for exploring the electronic alternative.

I am obliged to Mr. Barron for providing support for much of the paper. With regard to the questions posed, resistors and capacitors which are stable enough for electronic circuits are now obtainable. Valves and rectifiers are somewhat less certain but it is expected that the stability of all components will improve as time proceeds; frequency and voltage variations in the a.c. power supply are not a source of great difficulty in the small installations now being worked experimentally, and their effects can be reduced to any desired extent at a price; dry joints are a source of trouble, but it is too early yet for the value of printed circuits in electronic-exchange construction to be known, and I have no knowledge of a successful welding process to replace soldering in apparatus of this type.

Dr. Tillman's factual contribution is a valuable addition to the paper and also answers Mr. Hartley's point on the quality of time-division-multiplex transmission. With regard to Mr.

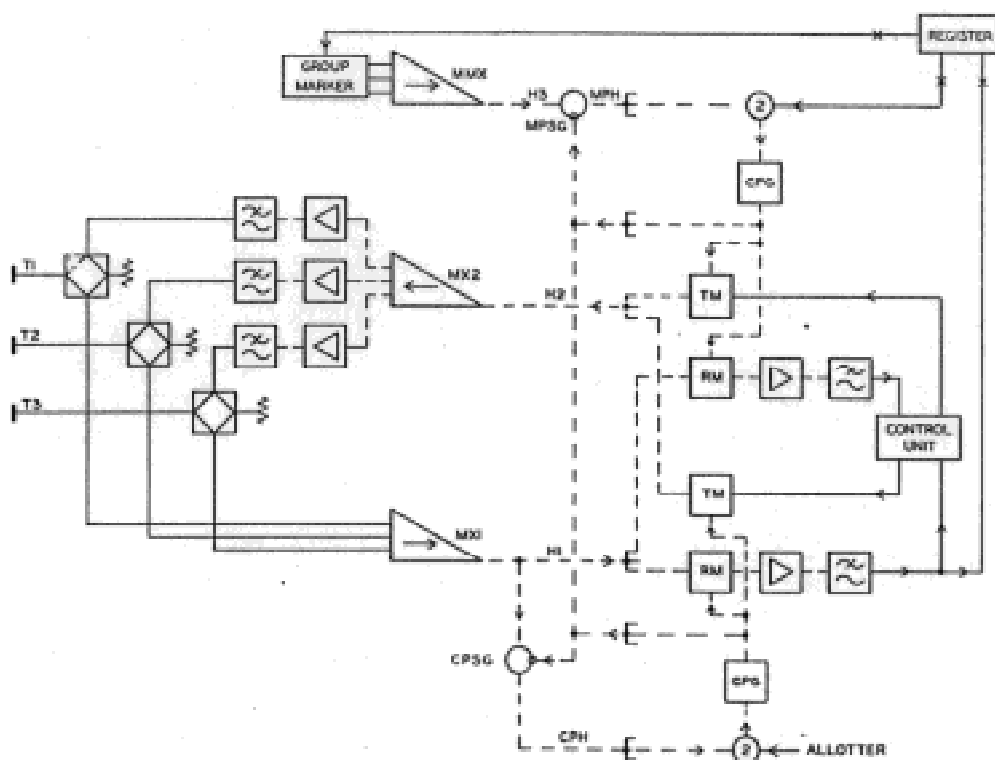


Fig. D.—Time-division-multiplex exchange.

Hartley's views on relays and subscriber's message-registers, I agree that the latter is the first thing that one would want to see replaced by an electronic device, but so far it is only the relay that has been shown to be easily and completely replaceable in this way.

Mr. Foot described some novel ideas, but I should need more details in order to comment on them. The subject of maintenance of electronic or any type of exchange is large enough for a paper by itself, and could be included in this one only in very general terms. I agree with Mr. Palmer's remarks on the subject and realize that an entirely new system offers great possibilities for incorporating maintenance as a positive and essential part of the design.

In reply to Mr. Davison, 4-wire switching is inherent in some electronic systems and the complication cannot then be avoided. The 4-wire trunking which results is an advantage to speech transmission which can be used if desired to cheapen line construction to offset the extra cost of the switches. With regard to the number of thermionic valves used, time-division multiplexing causes a large power loss which has to be made up by amplification; low power-handling capacity is characteristic of electronic connector switches of all types, owing to the apparatus used and not solely to the use of amplifiers. With regard to the last question, all the systems described in the paper can be

applied to exchanges of any size by utilizing a sufficient number of switching stages in series.

I agree with Mr. McGavin that deterioration of electronic switches is comparable with contact erosion in electro-mechanical switches. The important points are that electronic switches are not subject to faults due to atmospheric dust and corrosion and the deterioration of at least some of them may be made practically negligible over many years. My statements regarding the negligible effects of assembly wiring, shipping and installation were intended to apply to the components, e.g. valves, rectifiers and so forth. I think that Mr. McGavin is referring to layouts, which I agree are important, but they are not likely to be affected in the way he implies.

Mr. Thompson exaggerates my remarks concerning contact-making devices. I do not think they should be avoided at all costs, and dials and switch-hook contacts are not out of place as the primary control devices of an electronic exchange. With regard to his final point, it is not theoretically necessary to have modulation and demodulation at each switching stage, but the question is too involved for this discussion. On the other hand, there is no great difficulty in making modulation and demodulation sufficiently good for them to be carried out many times in one conversation without the quality of the transmission falling below acceptable limits.